#1 · Phlebotomy Procedure Specimen Handling and Processing

Why does phlebotomy procedure and venipuncture technique carry such heavy weight on the CPTG exam?

Answer

The majority of preventable errors affecting patient results occur in the pre-analytical phase, before a specimen ever reaches an analyzer. A hemolyzed sample, a clotted tube, or a mislabelled specimen can produce a result that looks normal while being clinically meaningless or dangerous, and these are steps the phlebotomist directly controls.

#2 · Phlebotomy Procedure Specimen Handling and Processing

What is the first working step of a venipuncture procedure?

Answer

Reviewing the test requisition. It confirms which tests were ordered, which tube types and additives are needed, how many tubes must be filled, the draw order, and any special handling requirement such as protection from light or keeping the specimen on ice.

#3 · Phlebotomy Procedure Specimen Handling and Processing

What can happen if a phlebotomist glances carelessly at a requisition?

Answer

They may gather the wrong tube colour, under-collect a tube that needs a specific fill volume for an accurate anticoagulant ratio, or miss a special handling requirement, such as light protection or cold storage, that the ordered test requires.

#4 · Phlebotomy Procedure Specimen Handling and Processing

List the equipment a phlebotomist assembles before approaching the patient.

Answer

Tourniquet, appropriate gauge needle or winged set, tube holder, the correct tubes in the correct quantities, alcohol swabs, gauze, bandage material, a sharps container within reach, and gloves. Gathering everything first avoids leaving a partially completed draw to fetch a forgotten supply.

#5 · Phlebotomy Procedure Specimen Handling and Processing

Why is assembling all supplies before the draw important beyond convenience?

Answer

Leaving mid-draw to retrieve a forgotten item introduces both delay and infection control risk, since the phlebotomist may need to remove gloves, touch other surfaces, and return to a partially exposed patient site.

#6 · Phlebotomy Procedure Specimen Handling and Processing

What is one of the most serious sequencing errors a phlebotomist can make?

Answer

Skipping ahead to site selection before patient identification has been fully completed and verified. Every subsequent step becomes worthless if the specimen ends up drawn from the wrong person.

#7 · Phlebotomy Procedure Specimen Handling and Processing

When should hand hygiene be performed relative to glove donning?

Answer

Hands should be washed or sanitized with an alcohol-based hand rub directly before approaching the patient and immediately before gloving, not several minutes earlier while other tasks were still being completed.

#8 · Phlebotomy Procedure Specimen Handling and Processing

When are gloves put on during the venipuncture sequence?

Answer

After patient identification is confirmed and just before tourniquet application and site selection begin. Gloves must be donned fresh for each patient and never reused between patients under any circumstance.

#9 · Phlebotomy Procedure Specimen Handling and Processing

Which veins of the antecubital fossa are the preferred sites for routine venipuncture?

Answer

The median cubital, cephalic, and basilic veins, principally because of their size, accessibility, and relatively stable position. The phlebotomist inspects both antecubital areas before choosing a site.

#10 · Phlebotomy Procedure Specimen Handling and Processing

Why is visual inspection alone not enough when selecting a venipuncture site?

Answer

Palpation is what actually confirms a vein is suitable. A healthy vein feels round, firm but resilient under the fingertip, and rebounds after gentle pressure is released, a sensation often described as bouncy or springy, which cannot be judged by sight alone.

#11 · Phlebotomy Procedure Specimen Handling and Processing

How does a sclerosed or thrombosed vein feel on palpation, and what usually causes it?

Answer

It feels hard and cord-like and lacks the normal rebound, often the result of repeated prior punctures in the same location. Drawing from such a vein yields poor flow and an unreliable specimen, so it should not be used.

#12 · Phlebotomy Procedure Specimen Handling and Processing

How can a phlebotomist distinguish an artery from a vein by palpation, and why does this matter?

Answer

An artery has a palpable pulse and feels more elastic under pressure. It must never be used for routine venipuncture, since arterial puncture carries a higher risk of excessive bleeding and hematoma formation and mistaking an artery for a vein is a genuine safety event.

#13 · Phlebotomy Procedure Specimen Handling and Processing

Why is the basilic vein generally a secondary choice behind the median cubital vein?

Answer

The basilic vein sits close to the brachial artery and the median nerve, so using it carries a higher risk of accidental arterial puncture or nerve involvement compared to the more centrally located median cubital vein.

#14 · Phlebotomy Procedure Specimen Handling and Processing

Where and why is the tourniquet placed before venipuncture?

Answer

Approximately 7 to 10 centimetres above the intended puncture site, positioned to restrict venous return without cutting off arterial flow, allowing the target vein to distend and become easier to palpate and enter.

#15 · Phlebotomy Procedure Specimen Handling and Processing

How should tourniquet tension be judged during application?

Answer

Firm enough to engorge the vein but never so tight that it causes pain or restricts arterial inflow. A radial pulse should remain palpable distal to the tourniquet at all times while it is in place.

#16 · Phlebotomy Procedure Specimen Handling and Processing

How long should a tourniquet remain in place according to CLSI-based best practice, and why?

Answer

No longer than one minute, because prolonged application causes hemoconcentration, a local pooling and concentration effect that can falsely elevate results for analytes such as potassium, total protein, and cellular blood counts.

#17 · Phlebotomy Procedure Specimen Handling and Processing

Define hemoconcentration and explain its cause.

Answer

Hemoconcentration is a local pooling and concentration effect in which water and small molecules shift out of the vessel while larger molecules, cells, and cell-bound analytes become falsely elevated in concentration. It results from prolonged tourniquet time restricting venous return.

#18 · Phlebotomy Procedure Specimen Handling and Processing

What should a phlebotomist do if a vein cannot be located and entered within the first minute of tourniquet application?

Answer

Release the tourniquet, give the patient a brief rest period, and reapply the tourniquet only once site selection is ready to proceed again, rather than leaving it in place while continuing to search or reposition the needle.

#19 · Phlebotomy Procedure Specimen Handling and Processing

Which antiseptic agent is standard for routine venipuncture, and how is it applied?

Answer

70% isopropyl alcohol, applied in a circular motion beginning at the intended puncture point and working outward toward the periphery, which prevents recontaminating the cleaned centre with organisms from the surrounding skin.

#20 · Phlebotomy Procedure Specimen Handling and Processing

Why must the antiseptic site air dry completely before needle insertion?

Answer

Air drying allows the alcohol enough contact time to achieve its antiseptic effect and prevents the sting and specimen contamination risk of puncturing through wet alcohol, which can also cause hemolysis in the sample and a burning sensation for the patient.

#21 · Phlebotomy Procedure Specimen Handling and Processing

Is it acceptable to blow on or fan an antiseptic site to speed up drying?

Answer

No. Blowing on the site, fanning it, or wiping it dry with gauze defeats the purpose of air drying and should never be done, since it can reintroduce contamination or fail to allow adequate antiseptic contact time.

#22 · Phlebotomy Procedure Specimen Handling and Processing

Why do blood cultures require a more rigorous antiseptic protocol than routine venipuncture?

Answer

Even a small number of skin surface organisms introduced into a culture bottle can produce a false positive result that leads to unnecessary antibiotic treatment, so blood cultures use a substantially more rigorous site preparation sequence than a routine draw.

#23 · Phlebotomy Procedure Specimen Handling and Processing

How is the vein anchored before needle insertion, and why?

Answer

The thumb of the non-dominant hand is positioned a few centimetres below the intended puncture site, pulling the skin taut. Anchoring prevents the vein from rolling or shifting away from the needle, one of the more common causes of a failed or painful attempt.

#24 · Phlebotomy Procedure Specimen Handling and Processing

What does it mean for a needle to be inserted bevel-up, and why is this done?

Answer

The angled opening of the needle faces upward toward the skin surface. This allows for a cleaner entry into the vessel wall and a more controlled, smaller point of penetration than inserting bevel-down.

#25 · Phlebotomy Procedure Specimen Handling and Processing

What insertion angle range is generally used for venipuncture, and how does it vary by vein depth?

Answer

Approximately 15 to 30 degrees relative to the skin surface. Shallower veins call for an angle closer to 15 degrees, while deeper veins may require an angle nearer 30 degrees.

#26 · Phlebotomy Procedure Specimen Handling and Processing

What happens if the venipuncture insertion angle is too steep or too shallow?

Answer

Too steep an angle risks passing through the back wall of the vein, while too shallow an angle may fail to penetrate the vessel at all, resulting in a failed or incomplete draw.

#27 · Phlebotomy Procedure Specimen Handling and Processing

How does blood collection work with the evacuated tube system?

Answer

A vacuum-sealed tube is pushed into a tube holder attached to the needle, and the internal vacuum draws blood automatically once the tube's rubber stopper is pierced by the interior needle. Each tube fills until the vacuum is exhausted, then is withdrawn and replaced with the next tube while the outer needle stays stationary.

#28 · Phlebotomy Procedure Specimen Handling and Processing

How is blood collected and transferred when a syringe is used instead of an evacuated tube system?

Answer

Blood is drawn manually by retracting the plunger at a controlled, steady rate matched to venous flow. Once collection is finished, the blood must be transferred into tubes using a specific transfer device that maintains each tube's vacuum draw rather than injecting blood forcefully, which would risk hemolysis.

#29 · Phlebotomy Procedure Specimen Handling and Processing

What is a winged, or butterfly, set and when is it typically used?

Answer

A short flexible needle attached to plastic wings and connecting tubing, typically used for smaller or more fragile veins, such as those found in pediatric or elderly patients, or for veins on the back of the hand. It can connect to either a tube holder or a syringe.

#30 · Phlebotomy Procedure Specimen Handling and Processing

Why must tubes be filled in a specific order of draw regardless of collection system?

Answer

Additive carryover between tubes can alter results if the sequence is not respected. This applies equally whether tubes are filled directly by an evacuated tube system or filled afterward from a syringe.

#31 · Phlebotomy Procedure Specimen Handling and Processing

When should the tourniquet be released relative to needle withdrawal?

Answer

The tourniquet should be released before the needle is withdrawn, ideally once the last tube begins filling or immediately after, and in any case no later than the one-minute mark to avoid hemoconcentration.

#32 · Phlebotomy Procedure Specimen Handling and Processing

Why does releasing the tourniquet before needle withdrawal reduce hematoma risk?

Answer

It reduces pressure within the vein, lowering the likelihood of hematoma formation once the needle is removed, since a vessel under continued tourniquet pressure tends to leak more blood into the surrounding tissue at the puncture site.

#33 · Phlebotomy Procedure Specimen Handling and Processing

Describe correct needle withdrawal technique and gauze placement.

Answer

Withdrawal should be a single smooth motion performed at the same angle the needle was inserted, and gauze should be placed over the puncture site with gentle pressure applied the instant the needle clears the skin, not before, since pressing down on a partially inserted needle causes pain and tissue damage.

#34 · Phlebotomy Procedure Specimen Handling and Processing

How is a patient positioned to help apply pressure after needle withdrawal?

Answer

The patient, if able, may be asked to hold light pressure on the gauze with the arm extended and elevated, while the phlebotomist's attention turns immediately to safe disposal of the needle.

#35 · Phlebotomy Procedure Specimen Handling and Processing

What is the rule for needle disposal after a draw?

Answer

The used needle must be disposed of directly and immediately into a sharps container, activating any safety shielding mechanism as part of that single motion. There must be no recapping, no manual manipulation, and no walking any distance while carrying an exposed needle.

#36 · Phlebotomy Procedure Specimen Handling and Processing

Why are sharps containers positioned within arm's reach of the draw site?

Answer

So that immediate disposal is always possible without the phlebotomist needing to travel with a contaminated sharp, since carrying an exposed needle any distance is one of the events this positioning is meant to prevent.

#37 · Phlebotomy Procedure Specimen Handling and Processing

During which activities do the majority of documented needlestick injuries in healthcare settings occur?

Answer

The overwhelming majority occur during recapping, disassembly, or transport of used needles rather than during the venipuncture itself, which is why immediate, no-recap disposal is treated as non-negotiable rather than a matter of individual judgment.

#38 · Phlebotomy Procedure Specimen Handling and Processing

What steps follow safe needle disposal, before the phlebotomist leaves the patient?

Answer

Checking the puncture site for continued bleeding, applying a fresh bandage once bleeding has stopped, and giving the patient brief post-draw instructions on bandage duration, avoiding heavy lifting, and reporting unusual bruising, swelling, or discomfort.

#39 · Phlebotomy Procedure Specimen Handling and Processing

Where must every tube be labelled in Canadian phlebotomy practice?

Answer

At the patient's bedside, before leaving that patient's side under any circumstance. Labelling must never be deferred to a workstation, hallway, or any location away from the patient.

#40 · Phlebotomy Procedure Specimen Handling and Processing

Why is bedside labelling one of the firmest safety rules in Canadian phlebotomy practice?

Answer

Deferring labelling introduces the possibility that tubes from two different patients drawn in sequence become confused or swapped, an error that can range from a mismatched transfusion to an incorrect treatment decision based on someone else's laboratory values.

#41 · Phlebotomy Procedure Specimen Handling and Processing

What elements must a complete tube label include?

Answer

The patient's full legal name, a unique identifying number such as a health record or medical record number, date of birth, the date and time of collection, and the initials or identifier of the phlebotomist who performed the draw.

#42 · Phlebotomy Procedure Specimen Handling and Processing

What additional identifier do many facilities require on a specimen label?

Answer

A second identifier consistent with the identification method used at the start of the encounter, in addition to the core elements of name, unique number, date of birth, collection date and time, and phlebotomist identifier.

#43 · Phlebotomy Procedure Specimen Handling and Processing

What is the last opportunity to prevent a misidentification error before a specimen leaves the room?

Answer

Labelling immediately at the bedside allows the phlebotomist to visually and verbally reconfirm the patient's identity one final time against the freshly labelled tube while the patient is still present to catch any discrepancy.

#44 · Phlebotomy Procedure Specimen Handling and Processing

Define hematoma and list its common causes during venipuncture.

Answer

A hematoma is a localized collection of blood beneath the skin causing swelling, discolouration, and tenderness. It most commonly results from the needle passing through the back wall of the vein, insufficient pressure applied after withdrawal, or the tourniquet remaining in place after needle removal.

#45 · Phlebotomy Procedure Specimen Handling and Processing

How can a phlebotomist help prevent hematoma formation?

Answer

Prevention centres on proper needle angle and depth, prompt tourniquet release before withdrawal, and adequate direct pressure held for a sufficient period after the draw is complete.

#46 · Phlebotomy Procedure Specimen Handling and Processing

What behaviour is strongly associated with causing nerve injury during venipuncture?

Answer

Probing or redirecting the needle blindly once it has been inserted, sometimes described informally as fishing for a vein. This risks contacting a nerve running close to the major veins of the antecubital area, particularly near the basilic vein.

#47 · Phlebotomy Procedure Specimen Handling and Processing

What symptoms can result from nerve injury during a venipuncture attempt?

Answer

A needle moved side to side beneath the skin in search of a vessel can cause sharp shooting pain, numbness, or longer-term nerve damage if it contacts a nerve running near the puncture site.

#48 · Phlebotomy Procedure Specimen Handling and Processing

What is the correct response if a vein is not entered on the first pass?

Answer

Withdraw the needle fully and reassess. The needle should never be redirected under the skin while still inserted, since that is exactly the fishing behaviour that risks nerve injury.

#49 · Phlebotomy Procedure Specimen Handling and Processing

Define syncope in the context of phlebotomy and identify its usual triggers.

Answer

Syncope is a vasovagal response causing sudden lightheadedness or fainting. It can occur before, during, or after venipuncture and is triggered by anxiety, pain, or the sight of blood rather than by blood loss itself in most cases.

#50 · Phlebotomy Procedure Specimen Handling and Processing

What warning signs suggest a patient may be experiencing syncope?

Answer

Pallor, sweating, and the patient reporting feeling faint or nauseated are warning signs that a vasovagal episode may be starting, and the phlebotomist should be ready to respond promptly.

#51 · Phlebotomy Procedure Specimen Handling and Processing

How should a phlebotomist manage a patient showing signs of syncope?

Answer

Stop the draw if it is still in progress, lower the patient's head or have them lie down, loosen tight clothing, and monitor closely until the episode resolves, calling for additional assistance if the patient loses consciousness or does not recover promptly.

#52 · Phlebotomy Procedure Specimen Handling and Processing

Why is hemoconcentration considered a particularly dangerous complication?

Answer

It can silently distort results without producing any visible sign at the puncture site, making disciplined tourniquet timing the primary defence against it rather than any corrective action taken after the fact.

#53 · Phlebotomy Procedure Specimen Handling and Processing

How many venipuncture attempts is a single phlebotomist generally limited to before escalating to another staff member?

Answer

General Canadian practice guidance limits a single phlebotomist to no more than two attempts on any one patient before involving another qualified staff member.

#54 · Phlebotomy Procedure Specimen Handling and Processing

Why does the two-attempt limit exist for venipuncture?

Answer

It protects the patient from repeated unnecessary trauma and protects specimen quality, since multiple failed attempts increase the risk of hemolysis, hematoma, and patient distress, while also protecting the phlebotomist from continuing a technically difficult draw beyond what is reasonable.

#55 · Phlebotomy Procedure Specimen Handling and Processing

What professional behaviours are expected when a phlebotomist reaches the two-attempt limit?

Answer

Recognizing the limit, communicating clearly with the patient about the need to involve a colleague, and handing off calmly rather than persisting out of frustration or a wish to avoid asking for help.

#56 · Phlebotomy Procedure Specimen Handling and Processing

Why do exam questions on the two-attempt rule often appear as judgment-call scenarios?

Answer

Because they test whether a candidate understands that correct technique includes knowing when to stop, not simply how to perform each mechanical step of the draw correctly.

#57 · Phlebotomy Procedure Specimen Handling and Processing

What sensation distinguishes a healthy, usable vein from other structures on palpation?

Answer

A healthy vein feels round, firm but resilient, and rebounds after gentle pressure is released, often described as bouncy or springy. This tactile rebound is the key feature that separates it from a sclerosed vein or an artery.

#58 · Phlebotomy Procedure Specimen Handling and Processing

What effect does prolonged tourniquet time have specifically on potassium, total protein, and cell count results?

Answer

Prolonged tourniquet application causes hemoconcentration, which can falsely elevate the concentration of these analytes because water and small molecules shift out of the vessel while larger molecules and cells become concentrated in the remaining plasma.

#59 · Phlebotomy Procedure Specimen Handling and Processing

Why must a radial pulse remain palpable distal to the tourniquet throughout the draw?

Answer

It confirms the tourniquet is restricting venous return without cutting off arterial flow. If the radial pulse disappears, the tourniquet is too tight and must be loosened before proceeding.

#60 · Phlebotomy Procedure Specimen Handling and Processing

What is the purpose of applying antiseptic in an outward circular motion from the puncture point?

Answer

Starting at the intended puncture point and working outward prevents recontaminating the cleaned centre with organisms from the surrounding, less-cleaned skin, keeping the actual insertion site as sterile as possible.

#61 · Phlebotomy Procedure Specimen Handling and Processing

What consequence can puncturing through wet alcohol have on the specimen and the patient?

Answer

It can cause hemolysis in the sample and a burning sensation for the patient, which is why the antiseptic site must be allowed to air dry completely rather than being wiped or blown dry before insertion.

#62 · Phlebotomy Procedure Specimen Handling and Processing

In the evacuated tube system, what determines when a given tube stops filling?

Answer

Each tube fills until its internal vacuum is exhausted. The phlebotomist then withdraws that tube and replaces it with the next tube in the sequence while the outer needle remains stationary in the vein.

#63 · Phlebotomy Procedure Specimen Handling and Processing

Why must blood from a syringe be transferred into tubes with a specific transfer device rather than injected directly?

Answer

A proper transfer device maintains the vacuum draw of each tube, filling it gently. Injecting blood forcefully from a syringe into a tube risks hemolysis of the sample.

#64 · Phlebotomy Procedure Specimen Handling and Processing

Which patient populations or sites commonly call for a winged, or butterfly, set instead of a straight needle?

Answer

Smaller or more fragile veins, such as those found in pediatric or elderly patients, or veins on the back of the hand, are commonly drawn using a winged set because of its short, flexible needle and stabilizing wings.

#65 · Phlebotomy Procedure Specimen Handling and Processing

What post-draw instructions should a patient receive after bandaging?

Answer

To leave the bandage in place for a specified period, to avoid heavy lifting with that arm for a short time, and to notify staff if unusual bruising, swelling, or discomfort develops after they leave.

#66 · Phlebotomy Procedure Specimen Handling and Processing

Before applying a fresh bandage, what must the phlebotomist confirm about the puncture site?

Answer

That bleeding has actually stopped. The site is checked for continued bleeding first, and the fresh bandage is applied only once bleeding has stopped, not immediately after the initial gauze pressure.

#67 · Phlebotomy Procedure Specimen Handling and Processing

Why is anchoring the vein performed a few centimetres below, rather than directly at, the puncture site?

Answer

Anchoring below the puncture site pulls the skin taut over the vein without the phlebotomist's fingers being in the way of needle insertion, while still preventing the vein from rolling or shifting at the moment of entry.

#68 · Phlebotomy Procedure Specimen Handling and Processing

What is the relationship between insertion angle and needle withdrawal angle?

Answer

Needle withdrawal should be performed at the same angle the needle was inserted, as a single smooth motion, rather than at a different angle that could cause additional tissue trauma on the way out.

#69 · Phlebotomy Procedure Specimen Handling and Processing

Why should a phlebotomist never remove a needle from a tube holder by hand after a draw?

Answer

Manual manipulation of a used needle is one of the actions strictly prohibited during disposal, since it increases the risk of an accidental needlestick injury; the needle and holder must go directly into the sharps container together.

#70 · Phlebotomy Procedure Specimen Handling and Processing

Summarize why the pre-analytical phase, covered in this chapter, matters more to result accuracy than most people assume.

Answer

Analytical instruments themselves are highly reliable, so a normal-looking printed result gives no warning that the underlying specimen was compromised. Errors like hemolysis, clotting, mislabelling, or drawing from the wrong patient all happen before analysis and can make an accurate-looking result clinically meaningless or dangerous.

#71 · Phlebotomy Procedure Specimen Handling and Processing

What is the correct technique for mixing an additive tube after collection?

Answer

The tube is gently inverted, turned completely upside down and back to upright in one smooth motion, immediately after collection. This allows the air bubble to travel the full length of the specimen and mix the additive evenly. Shaking, vortexing, or snapping the wrist is never acceptable, since vigorous agitation introduces foam and damages red cell membranes.

#72 · Phlebotomy Procedure Specimen Handling and Processing

How many inversions does a light blue top sodium citrate tube require?

Answer

Three to four gentle inversions. This distributes the anticoagulant without disturbing the precise nine to one blood-to-citrate ratio that coagulation testing depends on for an accurate result.

#73 · Phlebotomy Procedure Specimen Handling and Processing

How many inversions does a green top heparin tube require, and why?

Answer

Eight to ten inversions. Heparin must bind evenly throughout the specimen to prevent any localized clotting in the plasma chemistry sample.

#74 · Phlebotomy Procedure Specimen Handling and Processing

Why does a lavender top potassium EDTA tube need eight to ten inversions?

Answer

EDTA must chelate calcium throughout the entire specimen quickly enough to prevent platelet clumping and early clot formation. This tube is used for hematology testing, including the complete blood count, so any clumping compromises the cell counts.

#75 · Phlebotomy Procedure Specimen Handling and Processing

What additive does a grey top tube contain and how many inversions does it need?

Answer

A grey top tube contains sodium fluoride and potassium oxalate and typically receives roughly eight to ten inversions. This allows the fluoride to begin halting glycolysis uniformly throughout the specimen rather than in isolated patches, which is essential for accurate glucose and lactate testing.

#76 · Phlebotomy Procedure Specimen Handling and Processing

How many inversions does a gold top serum separator tube require?

Answer

Generally five to six inversions. This distributes the clot activator through the specimen and promotes complete, even clot formation before centrifugation.

#77 · Phlebotomy Procedure Specimen Handling and Processing

Does a plain red top tube with no additive need the same deliberate inversion protocol as an additive tube?

Answer

No. Since there is no anticoagulant or clot activator to distribute, a red top tube does not require the same deliberate inversion protocol, although many facilities still recommend a few gentle inversions to help the specimen clot naturally and evenly rather than pooling against one wall of the tube.

#78 · Phlebotomy Procedure Specimen Handling and Processing

What happens to an anticoagulated tube that is under-mixed?

Answer

Pockets of blood never contact the additive and begin to clot within minutes. In a lavender top tube, even microscopic clots produce falsely low platelet counts and can obstruct the small-bore probes on automated hematology analyzers, forcing a redraw.

#79 · Phlebotomy Procedure Specimen Handling and Processing

What is the consequence of under-mixing a citrate tube used for coagulation testing?

Answer

Micro-clots form and yield unreliable coagulation results, which could mislead a clinician managing a patient's anticoagulation therapy. The specimen must be rejected and redrawn.

#80 · Phlebotomy Procedure Specimen Handling and Processing

What happens to red blood cells when a tube is over-mixed?

Answer

Vigorous shaking mechanically damages the red blood cell membrane, releasing hemoglobin and intracellular contents into the surrounding plasma or serum. This produces hemolysis, which can be seen as a pink or red tinge in an otherwise straw-coloured specimen.

#81 · Phlebotomy Procedure Specimen Handling and Processing

How can using too small a needle gauge cause hemolysis?

Answer

A needle gauge that is too small for the vacuum strength of the collection tube forces red blood cells through an opening that is not wide enough to accommodate them without shearing stress, rupturing their membranes as they pass through.

#82 · Phlebotomy Procedure Specimen Handling and Processing

Why is forcing blood from a syringe into an evacuated tube a known cause of hemolysis?

Answer

Drawing blood with a syringe and then pushing it through the needle into an evacuated tube, rather than letting the tube fill naturally, applies shearing stress to the red cells similar to using an undersized needle, rupturing their membranes.

#83 · Phlebotomy Procedure Specimen Handling and Processing

How does leaving a tourniquet on too long contribute to hemolysis?

Answer

Leaving the tourniquet on longer than the recommended one minute causes hemoconcentration and increases the fragility of the red cells in the sampled vein, making them more prone to rupture during collection.

#84 · Phlebotomy Procedure Specimen Handling and Processing

Why should a phlebotomist avoid drawing from a site with an existing hematoma?

Answer

Blood at a hematoma site has already begun to break down outside the vessel, contaminating the sample with free hemoglobin before the needle even enters the vein and producing a hemolyzed specimen.

#85 · Phlebotomy Procedure Specimen Handling and Processing

Name three additional technique errors, beyond tourniquet time and hematoma sites, that can cause hemolysis.

Answer

Pulling back too forcefully on a syringe plunger, using an improperly sized needle for a difficult vein, and allowing blood to travel through excessively narrow winged infusion set tubing at high vacuum can all shear and rupture red blood cells.

#86 · Phlebotomy Procedure Specimen Handling and Processing

Why does hemolysis cause a falsely elevated potassium result?

Answer

Intracellular potassium concentration is many times higher than plasma potassium concentration, so when red cells rupture and spill their contents, the reported potassium value can rise into a dangerously high range that does not reflect the patient's actual status, potentially triggering unnecessary or harmful clinical intervention.

#87 · Phlebotomy Procedure Specimen Handling and Processing

Which two enzymes are falsely elevated by hemolysis because they are concentrated inside red blood cells?

Answer

Lactate dehydrogenase, abbreviated LDH, and aspartate aminotransferase, abbreviated AST, are both present at high concentration inside red blood cells and become falsely elevated when those cells rupture and release their contents into the specimen.

#88 · Phlebotomy Procedure Specimen Handling and Processing

What typically happens to a hemolyzed specimen submitted for potassium, LDH, or AST testing?

Answer

The laboratory will typically reject the specimen and request a redraw, since reporting a result from a hemolyzed sample for these analytes could be clinically misleading and lead to inappropriate treatment decisions.

#89 · Phlebotomy Procedure Specimen Handling and Processing

What is centrifugation and what is its purpose?

Answer

Centrifugation spins a tube at high speed so that the denser cellular elements move outward and settle at the bottom while the lighter liquid, either serum or plasma, rises to the top. This separation is required before many tests can be analyzed.

#90 · Phlebotomy Procedure Specimen Handling and Processing

Is a phlebotomy technician expected to memorize exact centrifuge speeds and times for every specimen type?

Answer

No. A phlebotomy technician is not expected to memorize exact revolutions per minute or exact timing for every specimen type, but must understand that centrifugation must always follow the laboratory's validated protocol for each tube type.

#91 · Phlebotomy Procedure Specimen Handling and Processing

What problem occurs if a serum tube is centrifuged before the clot has fully formed?

Answer

Spinning a tube before its clot has fully formed can contaminate the serum with fibrin strands, which causes problems downstream in testing and analyzer performance.

#92 · Phlebotomy Procedure Specimen Handling and Processing

How long does a serum tube without a clot activator typically need to clot at room temperature before centrifugation?

Answer

Often up to sixty minutes. A serum tube without a clot activator generally needs this longer clotting time at room temperature before it is safe to centrifuge, compared with roughly thirty minutes for a tube that contains a clot activator.

#93 · Phlebotomy Procedure Specimen Handling and Processing

What is barrier gel and where is it found in a collection tube?

Answer

Barrier gel has a specific gravity between that of the cells and that of the serum or plasma. During centrifugation it migrates to a position between the two layers, forming a physical barrier that keeps them permanently separated even after the tube stops spinning.

#94 · Phlebotomy Procedure Specimen Handling and Processing

Which two tube types rely on barrier gel technology?

Answer

Gold top serum separator tubes and light green top plasma separator tubes both rely on a barrier gel that separates the cellular and liquid portions of the specimen after centrifugation.

#95 · Phlebotomy Procedure Specimen Handling and Processing

What is the practical laboratory advantage of barrier gel beyond keeping layers separated?

Answer

Barrier gel prevents the cells from continuing to metabolize substances in the serum, such as glucose, after separation, and it allows the same primary tube to often be used directly on automated analyzers without transferring the serum to a separate container.

#96 · Phlebotomy Procedure Specimen Handling and Processing

Which analyte is the most frequently tested example of a light-sensitive substance?

Answer

Bilirubin. It degrades measurably when exposed to ambient room light and even more aggressively to direct sunlight or phototherapy lighting used in nurseries, so bilirubin specimens must be protected from light immediately after collection.

#97 · Phlebotomy Procedure Specimen Handling and Processing

How should a bilirubin or folate specimen be protected during transport?

Answer

The tube should be wrapped in aluminum foil or placed inside an amber transport bag immediately after collection and kept covered until the moment of analysis, since both bilirubin and folate degrade quickly when exposed to light.

#98 · Phlebotomy Procedure Specimen Handling and Processing

Which vitamin, besides bilirubin's testing needs, is notably light-sensitive?

Answer

Folate, also called vitamin B9, is notably light-sensitive and degrades quickly when exposed to light, so a folate specimen should be protected in foil or an amber bag without delay, the same as a bilirubin specimen.

#99 · Phlebotomy Procedure Specimen Handling and Processing

Why must ammonia specimens be transported on ice?

Answer

Ammonia levels in a specimen can rise rapidly at room temperature as cellular metabolism continues after collection, so ammonia specimens must be chilled on ice immediately and processed quickly to avoid a falsely elevated result.

#100 · Phlebotomy Procedure Specimen Handling and Processing

Why is lactic acid, like ammonia, transported on ice?

Answer

Lactic acid continues to be produced by ongoing glycolysis within the specimen unless it is rapidly cooled. Placing lactate specimens on ice immediately after collection slows this process and preserves an accurate result.

#101 · Phlebotomy Procedure Specimen Handling and Processing

Why are blood gas specimens chilled on ice?

Answer

Chilling a blood gas specimen on ice slows ongoing cellular respiration within the sample, so the measured oxygen and carbon dioxide values still reflect the patient's status at the moment of collection rather than the specimen's own metabolism afterward.

#102 · Phlebotomy Procedure Specimen Handling and Processing

What are cold agglutinins and cryoglobulins, and how must a specimen tested for them be handled?

Answer

Cold agglutinins and cryoglobulins are proteins that clump together or precipitate out of solution when the specimen temperature drops. A specimen collected to test for these substances must be kept warm, ideally at or near body temperature, from collection through transport, and delivered promptly.

#103 · Phlebotomy Procedure Specimen Handling and Processing

What would happen if a cold agglutinin specimen were placed on ice, the way an ammonia specimen is?

Answer

Placing a cold agglutinin or cryoglobulin specimen on ice would destroy the sample's diagnostic value, because it would cause exactly the clumping reaction the test is meant to measure only under controlled conditions rather than as an artifact of transport.

#104 · Phlebotomy Procedure Specimen Handling and Processing

What three handling categories must a phlebotomy technician correctly recognize for a given test order?

Answer

A technician must recognize whether a specimen is light-protected, ice-chilled, or warmth-protected, and apply the correct handling consistently, since applying the wrong category can destroy the specimen's diagnostic value.

#105 · Phlebotomy Procedure Specimen Handling and Processing

Why is a clotted specimen in an anticoagulant tube unacceptable to the laboratory?

Answer

The presence of any clot means the anticoagulant failed to do its job, whether from under-mixing, a difficult or slow draw, or a defective tube. Results generated from such a specimen cannot be trusted, so the specimen is rejected.

#106 · Phlebotomy Procedure Specimen Handling and Processing

What does QNS mean and why is it especially critical for a citrate tube?

Answer

QNS means quantity not sufficient, a tube underfilled relative to its designed draw volume. It is especially critical for the light blue citrate tube because that tube is manufactured for a fixed nine to one ratio of blood to liquid sodium citrate, and underfilling leaves proportionally too much anticoagulant.

#107 · Phlebotomy Procedure Specimen Handling and Processing

What clinical effect does an underfilled citrate tube have on coagulation results?

Answer

The excess anticoagulant relative to blood artificially prolongs clotting times measured by the laboratory, which can lead to an inaccurate assessment of the patient's actual coagulation status.

#108 · Phlebotomy Procedure Specimen Handling and Processing

Is a hemolyzed specimen always rejected outright by the laboratory?

Answer

Not necessarily. A hemolyzed specimen is unacceptable for potassium, LDH, AST, and several other analytes, but it may still be usable for tests unaffected by hemolysis, depending on laboratory policy.

#109 · Phlebotomy Procedure Specimen Handling and Processing

According to Canadian laboratory practice, what must happen to a specimen that arrives unlabeled or with a label that does not match the requisition?

Answer

The specimen must be rejected and the patient must be redrawn. It is never acceptable to relabel a specimen after the fact based on the phlebotomist's memory of which patient it came from, no matter how confident the phlebotomist feels.

#110 · Phlebotomy Procedure Specimen Handling and Processing

Why is relabeling a specimen from memory never acceptable, even when the phlebotomist is confident?

Answer

Once a specimen leaves the patient's presence unlabeled, there is no reliable way to correlate it back to the correct individual with the certainty patient safety demands. A mislabeled specimen reported under the wrong name could cause a transfusion reaction, an incorrect diagnosis, or a wrong treatment decision.

#111 · Phlebotomy Procedure Specimen Handling and Processing

When must positive patient identification and label verification occur relative to collection?

Answer

Every accredited laboratory requires positive patient identification and label verification to occur at the bedside, immediately after collection and before the phlebotomist leaves the patient, to prevent labeling errors from ever having the chance to occur.

#112 · Phlebotomy Procedure Specimen Handling and Processing

What is chain of custody and when is it required?

Answer

Chain of custody is an unbroken, fully documented record of every person who has handled a specimen from collection through final disposal or storage. It applies to specimens collected for legal or forensic purposes, most commonly certain drug and alcohol testing for employment, insurance, or medico-legal reasons.

#113 · Phlebotomy Procedure Specimen Handling and Processing

How is a forensic or legal specimen sealed and documented at the moment of collection?

Answer

It is typically collected using a specialized kit, sealed with tamper-evident tape immediately in the presence of the donor, and accompanied by a chain of custody form that both the donor and the collector sign and date at the time of collection.

#114 · Phlebotomy Procedure Specimen Handling and Processing

What must be documented at every subsequent transfer of custody for a legal or forensic specimen?

Answer

Every transfer, whether to a courier, a laboratory receiving technician, or a testing analyst, must be documented with a signature, a date, and a time, so a complete and verifiable record exists if the result is ever challenged in a legal proceeding.

#115 · Phlebotomy Procedure Specimen Handling and Processing

What is the consequence of even a minor break in the documented chain of custody?

Answer

Even a minor break in the documented chain can render the result inadmissible in a legal proceeding, which is why a phlebotomy technician must recognize a legal or forensic order and follow the stricter protocol precisely.

#116 · Phlebotomy Procedure Specimen Handling and Processing

Do routine clinical specimens require the same chain of custody documentation as forensic specimens?

Answer

No. Routine clinical specimens do not require this level of documentation, but a phlebotomy technician must still recognize when an order specifically calls for a legal or forensic collection and switch to the stricter protocol.

#117 · Phlebotomy Procedure Specimen Handling and Processing

What general principle applies to how long a properly collected specimen remains a valid, testable sample?

Answer

Transport and processing delays are not neutral. Some analytes are stable at room temperature for many hours, while others begin to degrade within minutes, so a specimen sitting in a collection tray longer than necessary is quietly becoming less reliable.

#118 · Phlebotomy Procedure Specimen Handling and Processing

For a timed specimen, which collection time should be recorded, the ordered time or the actual time?

Answer

The actual collection time must be recorded, not the ordered or scheduled time. This allows the physician interpreting the result, such as a medication level drawn at a specific interval after a dose, to correctly account for how much time had truly elapsed.

#119 · Phlebotomy Procedure Specimen Handling and Processing

What should a phlebotomist do if a patient reports having eaten within the required fasting window before a fasting glucose test?

Answer

That fact must be documented and communicated, and the specimen should be flagged as non-fasting on the requisition or specimen label, since silently processing it as though the fasting requirement was met risks a misleading result affecting a diabetes diagnosis or cholesterol treatment decision.

#120 · Phlebotomy Procedure Specimen Handling and Processing

What is point-of-care testing, abbreviated POCT?

Answer

POCT refers to testing performed at or near the patient's location rather than sending the specimen to a central laboratory. Common examples include capillary glucose testing at the bedside, rapid strep testing, and portable blood gas analyzers used in emergency and critical care settings.

#121 · Phlebotomy Procedure Specimen Handling and Processing

Why are transport, centrifugation, and stability concerns less relevant for POCT specimens?

Answer

POCT specimens are typically analyzed within minutes of collection using small, often single-use devices, so there is essentially no delay between collection and analysis, unlike specimens sent to a central laboratory.

#122 · Phlebotomy Procedure Specimen Handling and Processing

Does POCT eliminate the risk of collection-related errors affecting a result?

Answer

No. POCT still demands correct technique at the collection step, proper quality control on the device, and accurate documentation of results. Squeezing a capillary puncture site too hard can dilute the sample with tissue fluid and produce a misleading POCT result.

#123 · Phlebotomy Procedure Specimen Handling and Processing

What is a delta check?

Answer

A delta check compares a patient's new test result against that same patient's own previous result for the same analyte, looking for a change larger than what would be physiologically plausible over the time interval between the two draws.

#124 · Phlebotomy Procedure Specimen Handling and Processing

When a delta check flags an unexpectedly large swing in a result, what does the laboratory consider first?

Answer

The laboratory does not assume the patient's condition changed dramatically without first considering whether something went wrong before the specimen reached the analyzer, such as a mislabeled tube, a hemolyzed sample, or a specimen drawn from the wrong patient entirely.

#125 · Phlebotomy Procedure Specimen Handling and Processing

Does a phlebotomy technician need detailed statistical knowledge of how delta checks are calculated?

Answer

No. A phlebotomy technician does not perform delta checks and does not need detailed statistical knowledge of the calculation, but understanding that this safeguard exists reinforces why correct mixing, protection of light- and temperature-sensitive specimens, accurate labeling, and honest documentation all matter.

#126 · Tubes Colors and Additives

What organization's colour-coding framework governs tube stopper colours across Canadian laboratories?

Answer

The Clinical and Laboratory Standards Institute (CLSI) framework governs tube colour-coding across Canadian laboratory practice. This standardization means a lavender top from one manufacturer behaves identically to a lavender top from another. It lets a CPT-certified technician move between Canadian sites without relearning the additive system.

#127 · Tubes Colors and Additives

What additive is inside a blood culture bottle, and what is its abbreviation?

Answer

Blood culture bottles contain sodium polyanetholesulfonate, abbreviated SPS, dissolved in a nutrient broth designed to support microbial growth. SPS is not a simple anticoagulant; its main job is protecting organisms in the specimen so they can be recovered and identified on incubation.

#128 · Tubes Colors and Additives

Name the three ways SPS helps a blood culture bottle recover organisms successfully.

Answer

SPS works by inhibiting complement proteins, inhibiting phagocytic white blood cells, and neutralizing certain antibiotics already circulating in the patient's blood from prior treatment. Together these actions stop the body's own defences and residual drugs from killing the very organisms the physician needs identified.

#129 · Tubes Colors and Additives

Why do blood culture bottles need growth promotion rather than only clot prevention?

Answer

A normal draw's immune defences and any residual antibiotic would work against recovering bacteria or fungi, actively suppressing organisms present in the specimen. SPS interrupts that defensive process just enough that any microorganisms present can survive and multiply once the bottle is incubated, which is exactly what a blood culture is designed to detect.

#130 · Tubes Colors and Additives

Why must blood culture bottle tops receive a distinct antiseptic protocol from routine venipuncture site prep?

Answer

Any skin flora introduced into the bottle will grow just as readily as a true pathogen once incubated, producing a false positive that can lead to unnecessary treatment. Because of this risk, each bottle top, aerobic or anaerobic, requires its own dedicated antiseptic protocol separate from ordinary site prep.

#131 · Tubes Colors and Additives

What three components make up the acid citrate dextrose (ACD) additive?

Answer

ACD combines citric acid, sodium citrate, and dextrose in one formulation. This mix preserves red blood cell antigens and DNA integrity over longer storage periods than most other anticoagulants allow, which is why ACD is chosen for specialized testing that depends on cell and DNA stability over time.

#132 · Tubes Colors and Additives

A yellow-topped ACD tube and a blood culture bottle can share a similar colour. Why must a technician never treat them as the same tube?

Answer

Their purposes are entirely different despite the shared colour in some systems. Blood culture bottles use SPS in a nutrient broth to grow organisms, while ACD tubes use citric acid, sodium citrate, and dextrose to preserve cell antigens and DNA for specialized testing, so confusing the two by colour alone is a testable error.

#133 · Tubes Colors and Additives

Name three specialized applications for which ACD tubes are the preferred draw.

Answer

ACD tubes are preferred for blood bank compatibility testing when cellular integrity over time matters, for DNA-based paternity or parentage testing where genetic material must remain undamaged, and for HLA phenotyping used in transplant matching and certain immunological workups.

#134 · Tubes Colors and Additives

What role does the dextrose component play in an ACD tube?

Answer

The dextrose component provides an energy source that helps keep red blood cells viable for specialized downstream applications like blood bank, DNA, and HLA testing. A simple citrate tube without the added sugar cannot offer this same degree of cell viability over time.

#135 · Tubes Colors and Additives

What additive is in the light blue top tube?

Answer

The light blue top tube is built around sodium citrate. It is one of the most tightly regulated tubes on the tray because coagulation testing is extremely sensitive to any collection error affecting the citrate-to-blood ratio.

#136 · Tubes Colors and Additives

By what mechanism does sodium citrate prevent clotting in a light blue top tube?

Answer

Sodium citrate prevents clotting by chelating calcium, meaning it binds free calcium ions in the plasma and removes them from availability. Calcium is an essential cofactor at several steps of the coagulation cascade, so without free calcium the cascade cannot proceed to fibrin formation.

#137 · Tubes Colors and Additives

Why is sodium citrate's calcium chelation described as reversible, and why does that matter?

Answer

Sodium citrate's chelation is reversible, which is what makes the light blue tube useful for coagulation testing at all. When the laboratory adds calcium back to the specimen along with specific reagents, the clotting reaction can be triggered and timed under controlled conditions, forming the basis of PT/INR and PTT/aPTT testing.

#138 · Tubes Colors and Additives

What do the abbreviations PT/INR and PTT/aPTT stand for, and what do they measure?

Answer

PT/INR is prothrombin time with international normalized ratio, and PTT or aPTT is activated partial thromboplastin time. Both measure how long it takes plasma to clot once the light blue tube's inhibiting calcium chelation has been reversed by the laboratory adding calcium back in.

#139 · Tubes Colors and Additives

What is the required blood-to-anticoagulant ratio for a light blue top citrate tube?

Answer

The light blue top is manufactured to a strict nine to one ratio of blood to anticoagulant. The entire coagulation test depends on this precise, known ratio, which is why the tube must be filled exactly to its fill line on every draw.

#140 · Tubes Colors and Additives

What happens to a coagulation result if a light blue top tube is underfilled?

Answer

An underfilled or QNS (quantity not sufficient) light blue tube has proportionally too much citrate relative to the blood collected. That excess citrate binds more calcium than the standard ratio anticipates, so the laboratory's fixed, standardized calcium reagent cannot fully reverse the chelation, producing an artificially prolonged clotting time.

#141 · Tubes Colors and Additives

Why do laboratories reject underfilled coagulation tubes instead of testing them?

Answer

An underfilled light blue tube produces an artificially prolonged clotting time that reflects a collection error rather than the patient's true coagulation status. Because the result would be misleading, laboratories reject these specimens rather than reporting an inaccurate value.

#142 · Tubes Colors and Additives

How can an unusually high hematocrit distort a light blue tube result even at a full draw?

Answer

A high hematocrit means a smaller plasma fraction relative to red cell volume, which increases the effective concentration of citrate in the available plasma. This produces the same effect as underfilling: the laboratory's fixed calcium reagent cannot fully reverse the excess chelation, giving a falsely prolonged result even though the tube was filled completely to the line.

#143 · Tubes Colors and Additives

What additive, if any, is inside a plain red top tube?

Answer

A plain red top tube, sometimes called a non-additive tube, contains no anticoagulant and no clot activator at all. Blood placed in it clots naturally through its own intrinsic and extrinsic pathways, and after centrifugation the laboratory recovers serum.

#144 · Tubes Colors and Additives

What is serum, as recovered from a plain red top tube?

Answer

Serum is plasma with the clotting factors and fibrinogen already consumed in the clot. It is what remains after whole blood clots naturally and is centrifuged to separate the liquid fraction from the clot and cells.

#145 · Tubes Colors and Additives

Why is a plain red top chosen for certain serology, blood bank, and specialized chemistry assays over an SST?

Answer

A plain red top has no additive whatsoever, so it is chosen when a test result could be affected by any additive residue, including gel. For assays where absolute additive purity matters more than convenience, a plain red top eliminates that risk entirely.

#146 · Tubes Colors and Additives

What is the tradeoff of using a plain red top tube instead of an SST?

Answer

The tradeoff is time. A plain red top can take significantly longer to fully clot than a tube containing a clot activator, because it has nothing accelerating the intrinsic clotting pathway, which matters when turnaround time is clinically urgent.

#147 · Tubes Colors and Additives

What two components are inside a gold top serum separator tube (SST)?

Answer

An SST contains a clot activator, typically fine silica particulate, that speeds clotting, and a thixotropic separator gel with a density engineered to sit between packed red cells and serum after centrifugation. These two components do very different jobs at different stages of processing.

#148 · Tubes Colors and Additives

How does the silica clot activator in an SST speed up clot formation?

Answer

The fine silica particulate provides a large surface area that accelerates the intrinsic clotting pathway, shortening the time to full clot formation compared to a plain red top. This addresses the turnaround-time tradeoff a non-additive red top carries.

#149 · Tubes Colors and Additives

What happens to the separator gel in an SST during centrifugation?

Answer

The gel migrates to a position between the density of packed red cells and the density of serum. This forms a physical barrier that keeps the serum on top cleanly separated from the cellular components below, allowing serum to be poured or aspirated without contamination.

#150 · Tubes Colors and Additives

Why can an SST specimen be safely stored and revisited over time, unlike a plain red top?

Answer

The gel barrier prevents red cells from leaching intracellular contents, such as potassium, back into the serum over time. This physical separation is what makes SST specimens suitable for routine chemistry, serology, and immunology panels even after some storage delay.

#151 · Tubes Colors and Additives

Does the separator gel in an SST prevent clotting? Explain.

Answer

No. The gel does nothing to prevent clotting itself; the gold top tube clots exactly as a plain red top would, only faster because of its silica clot activator. The gel's entire function is to physically separate serum from clot debris and cells after clotting has already happened.

#152 · Tubes Colors and Additives

This is one of the most commonly tested concepts on the CPT exam regarding SST tubes. What is it?

Answer

The distinction between a mechanical separator and a true anticoagulant. The SST gel is a physical barrier used only after centrifugation, not an anticoagulant or clot inhibitor, and treating it as though it prevents clotting is a documented misunderstanding of how the tube works.

#153 · Tubes Colors and Additives

What additive class does the green top tube introduce, and what forms can it take?

Answer

The green top tube contains heparin, a genuinely different class of anticoagulant from citrate or EDTA. It is available as sodium heparin, lithium heparin, or ammonium heparin depending on the manufacturer and the specific downstream test being run.

#154 · Tubes Colors and Additives

Does heparin chelate calcium the way citrate and EDTA do?

Answer

No. Heparin does not chelate calcium at all, which is the key mechanistic difference between heparin and both citrate and EDTA. Heparin achieves its anticoagulant effect through an entirely separate pathway.

#155 · Tubes Colors and Additives

By what mechanism does heparin actually prevent clotting?

Answer

Heparin binds to and dramatically potentiates antithrombin III, a naturally occurring plasma protein that inhibits thrombin and several other activated clotting factors. Once heparin binds antithrombin III, the rate at which it inactivates thrombin and factor Xa increases enormously, shutting down the cascade downstream of calcium.

#156 · Tubes Colors and Additives

What does antithrombin III do in the absence of heparin, compared to when heparin is present?

Answer

In the absence of heparin, antithrombin III neutralizes clotting factors slowly on its own. Once heparin binds it, the rate at which antithrombin III inactivates thrombin and factor Xa increases enormously, which is how heparin exerts its powerful anticoagulant effect.

#157 · Tubes Colors and Additives

Why does blood in a green top tube become plasma rather than serum?

Answer

Heparin prevents clotting by potentiating antithrombin III, leaving calcium untouched but shutting down the coagulation cascade downstream. Because the blood never clots, the laboratory recovers plasma rather than the serum obtained from a naturally clotted specimen.

#158 · Tubes Colors and Additives

Why are green top heparin tubes commonly used for stat chemistry panels?

Answer

Heparin tubes give the laboratory plasma quickly, since there is no waiting for clot formation at all as there would be with a serum tube. This makes green top tubes useful whenever turnaround time is critical, such as in stat chemistry testing.

#159 · Tubes Colors and Additives

Why is a heparin tube specifically favoured for ammonia level testing?

Answer

Ammonia testing requires an anticoagulant that does not itself introduce chemical interference, and heparin fits that requirement. This is one of the specialized tests, beyond routine stat chemistry, for which the green top's heparin chemistry is specifically chosen.

#160 · Tubes Colors and Additives

What is a plasma separator tube (PST), and what does it combine?

Answer

A PST combines lithium heparin with a separator gel functioning much like the SST's gel. It gives laboratories a fast, gel-separated plasma specimen, which is useful for high-volume stat chemistry testing where both speed and clean separation are needed.

#161 · Tubes Colors and Additives

What additive is inside a lavender or purple top tube, and what is its abbreviation?

Answer

The lavender top tube contains ethylenediaminetetraacetic acid, universally abbreviated EDTA. It is the standard tube for hematology testing on the phlebotomy tray.

#162 · Tubes Colors and Additives

How does EDTA's calcium chelation differ from sodium citrate's chelation?

Answer

Like citrate, EDTA works by chelating calcium, but unlike citrate's reversible binding, EDTA's chelation is essentially irreversible under normal laboratory conditions. This single difference explains why the two tubes are used for entirely different categories of testing.

#163 · Tubes Colors and Additives

Why is EDTA an excellent anticoagulant for preserving whole blood cellular morphology?

Answer

EDTA prevents clotting completely and does so without causing the cell shrinkage or swelling artifacts that some other additives can introduce. Red blood cell size, white blood cell morphology, and platelet appearance are all best preserved in EDTA-anticoagulated blood.

#164 · Tubes Colors and Additives

Name four hematology applications for which EDTA is the anticoagulant of choice.

Answer

EDTA is the anticoagulant of choice for a complete blood count (CBC), differential counts, reticulocyte counts, and peripheral blood smear preparation. In each case cellular morphology needs to be preserved accurately, which EDTA does better than other additives.

#165 · Tubes Colors and Additives

Is EDTA-anticoagulated blood technically a plasma or a serum specimen?

Answer

Neither in the strict sense. EDTA plasma is technically an anticoagulated whole blood specimen rather than a true plasma or serum, since it has not been centrifuged to remove cells and has not clotted to consume clotting factors.

#166 · Tubes Colors and Additives

Why is EDTA completely unsuitable for coagulation testing?

Answer

Because EDTA's calcium chelation is irreversible, the laboratory cannot add calcium back to reverse the effect the way it can with citrate. Any attempt to run a PT or PTT on EDTA-anticoagulated blood would produce meaningless results.

#167 · Tubes Colors and Additives

Why is EDTA also unsuitable for many chemistry tests?

Answer

EDTA's irreversibly bound calcium makes any calcium level reported from that tube inaccurate, and EDTA itself can interfere with numerous enzymatic assays and electrolyte measurements. These two problems make lavender top tubes a poor substitute for chemistry draws.

#168 · Tubes Colors and Additives

A technician assumes a lavender top can substitute for a light blue or gold top because all three involve some form of anticoagulation. What is wrong with this reasoning?

Answer

The underlying chemistry involved is fundamentally different between the three tubes, and the downstream tests each one supports do not overlap. A lavender top's irreversible EDTA chelation cannot be reversed for coagulation testing, and EDTA interferes with many chemistry assays, so substitution produces invalid results.

#169 · Tubes Colors and Additives

What two components does a grey top tube contain?

Answer

Grey top tubes contain sodium fluoride combined with either potassium oxalate or EDTA. Each component plays a distinct role, so it is worth being precise about which one does which job.

#170 · Tubes Colors and Additives

Which component of a grey top tube actually provides the anticoagulant activity?

Answer

The actual anticoagulant activity in a grey top comes from the potassium oxalate or EDTA, which chelates calcium in essentially the same way described for citrate and EDTA tubes elsewhere. Sodium fluoride itself does not prevent clotting.

#171 · Tubes Colors and Additives

What is the role of sodium fluoride in a grey top tube?

Answer

Sodium fluoride is an antiglycolytic agent, meaning it inhibits the enzyme enolase in the glycolytic pathway that red blood cells and any bacteria present use to metabolize glucose. Its job is metabolic preservation, not clot prevention.

#172 · Tubes Colors and Additives

What happens to a glucose result if red blood cells are allowed to keep metabolizing glucose after collection?

Answer

Red blood cells continue consuming glucose from the plasma after collection at roughly ten percent per hour at room temperature. This would cause a glucose result to fall artificially low the longer a specimen sits before testing, particularly when there is any delay reaching the laboratory.

#173 · Tubes Colors and Additives

Why are grey top tubes the standard choice for glucose testing when transport delay is expected?

Answer

Sodium fluoride essentially freezes glucose metabolism in place at the moment of collection by inhibiting enolase in the glycolytic pathway. This prevents the roughly ten percent per hour glucose drift that would otherwise occur, protecting the accuracy of a delayed glucose result.

#174 · Tubes Colors and Additives

Besides glucose, what other test commonly uses a grey top tube, and why?

Answer

Grey top tubes are also standard for lactate testing, because the same rapid metabolic drift that distorts glucose would otherwise distort lactate results too. Sodium fluoride's antiglycolytic effect protects both analytes from post-collection metabolism.

#175 · Tubes Colors and Additives

Why does a grey top tube always contain both sodium fluoride and an anticoagulant together?

Answer

Sodium fluoride alone does not prevent clotting, so it must be paired with an actual anticoagulant such as potassium oxalate or EDTA. Without that pairing, the specimen would clot even though glucose metabolism was inhibited.

#176 · Tubes Colors and Additives

What defines a royal blue or dark blue top tube, if it is not a single specific additive chemistry?

Answer

Royal blue top tubes are defined less by a single additive and more by the manufacturing standard behind the tube itself. They are produced with rigorous controls to minimize trace metal contamination from the manufacturing process.

#177 · Tubes Colors and Additives

Why does trace metal contamination matter so much for a royal blue top tube?

Answer

Even microscopic amounts of lead, zinc, copper, or other trace elements leaching from ordinary glass, stopper rubber, or lubricant could meaningfully distort a trace element or toxicology result measured in parts per billion. Royal blue tubes are manufactured to avoid contributing any of that contamination.

#178 · Tubes Colors and Additives

Do all royal blue top tubes contain the same internal additive?

Answer

No. Because the defining feature is metal purity rather than a specific anticoagulant, royal blue tops are manufactured in more than one internal configuration: some are plain with no additive, functioning like a trace-element-safe red top, while others contain EDTA or heparin.

#179 · Tubes Colors and Additives

What must a technician check before drawing a royal blue top tube, and why is assuming interchangeability a testable error?

Answer

A technician must check the requisition or laboratory reference manual to confirm which internal additive, if any, that draw requires. Assuming all royal blue tubes are interchangeable is a common and testable error, since some contain no additive and others contain EDTA or heparin depending on whether serum or plasma is required.

#180 · Tubes Colors and Additives

Name three types of testing for which royal blue top tubes are most often ordered.

Answer

Royal blue top tubes are most often ordered for lead level testing, other trace metals such as zinc and copper, and toxicology panels. In each case accuracy depends on the tube contributing nothing measurable to the specimen.

#181 · Tubes Colors and Additives

Why does fill volume matter for every additive tube, not only the light blue citrate tube?

Answer

Every additive is manufactured in a fixed quantity calibrated to an expected blood volume, so underfilling any additive tube skews the additive-to-specimen ratio. This principle applies broadly across the whole tube system, even though the light blue tube enforces it most strictly.

#182 · Tubes Colors and Additives

What can happen to red blood cells in an underfilled EDTA tube?

Answer

An underfilled EDTA tube concentrates the anticoagulant relative to the blood present, which can shrink red blood cells through osmotic effects. This distorts cell counts and morphology on a CBC, even though EDTA is normally chosen specifically to preserve cellular appearance.

#183 · Tubes Colors and Additives

Does underfilling a heparin or fluoride tube still prevent clotting, and what problem remains?

Answer

An underfilled heparin or fluoride tube can still prevent clotting, but it skews the additive-to-specimen ratio even so. This concentration problem can still affect downstream results, even though clot prevention itself is not the failure.

#184 · Tubes Colors and Additives

Which tube's fill ratio is the most sensitive to underfilling and the most strictly enforced by rejection criteria?

Answer

The light blue top's nine to one ratio is the most sensitive to underfilling and the most strictly enforced by laboratory rejection criteria of any tube on the tray. Even so, the general principle that concentration matters applies to every additive tube, not just this one.

#185 · Tubes Colors and Additives

Why does insufficient mixing risk microclots in an anticoagulant tube like light blue, green, lavender, or grey?

Answer

Inversion brings the additive into contact with the full blood volume before clotting factors can begin their work. Insufficient mixing risks partial clot formation, sometimes called microclots, which can clog analyzers or invalidate results.

#186 · Tubes Colors and Additives

Why does mixing matter for a gold top SST, if the tube is designed to clot anyway?

Answer

For gold top SST tubes, adequate mixing activates the clot activator evenly throughout the specimen so the entire sample clots at a consistent rate. Without even mixing, the specimen can clot unevenly and trap serum within a partially formed clot.

#187 · Tubes Colors and Additives

Why does an expired tube matter even if the technician's venipuncture technique is flawless?

Answer

A tube's expiration date reflects the real chemical stability of the additive inside, not an arbitrary formality. A technician who uses an expired tube risks a specimen that looks properly collected but produces inaccurate results for reasons unrelated to venipuncture skill.

#188 · Tubes Colors and Additives

How can an expired liquid anticoagulant tube like a light blue citrate tube fail even with correct draw technique?

Answer

Liquid anticoagulants such as sodium citrate can slowly lose concentration or shift pH over time even in a sealed, properly stored tube. The vacuum draw itself can also degrade with age, resulting in a short draw that fails to reach the fill line even with correct technique.

#189 · Tubes Colors and Additives

When should a technician check a tube's expiration date relative to the draw?

Answer

Checking the expiration date is one of the pre-collection steps performed on every tube before it touches a patient. This applies to every tube type, not just those with liquid anticoagulants, since vacuum integrity can also degrade with age.

#190 · Tubes Colors and Additives

What is the most common CPT exam trap involving the SST's separator gel?

Answer

The most common trap is treating the separator gel as though it were itself an anticoagulant. It is neither; the gel does nothing until after centrifugation, and a candidate who describes it as preventing clotting has misunderstood the tube completely.

#191 · Tubes Colors and Additives

What is the second recurring exam trap described in this chapter?

Answer

The second recurring trap is confusing heparin's mechanism with that of citrate or EDTA. All three additives stop a sample from clotting, but heparin potentiates antithrombin III to inactivate thrombin and other clotting factors, while citrate and EDTA chelate calcium away from the cascade entirely.

#192 · Tubes Colors and Additives

If an exam question asks which additive works by binding calcium, which additive must be correctly excluded?

Answer

Heparin must be excluded, since it works by potentiating antithrombin III rather than by chelating calcium. Citrate and EDTA are the two additives that actually achieve their anticoagulant effect by binding free calcium.

#193 · Tubes Colors and Additives

If an exam question asks which additive potentiates antithrombin III, which additives must be correctly excluded?

Answer

Citrate and EDTA must be excluded, since both achieve their anticoagulant effect by chelating calcium rather than by potentiating antithrombin III. Heparin is the additive that binds and dramatically increases antithrombin III's activity.

#194 · Tubes Colors and Additives

What four broad categories do tube additives fall into?

Answer

Additives fall into anticoagulants that prevent clotting through different mechanisms, clot activators that speed clotting along, antiglycolytic agents that stop cellular metabolism, and separator gels that create a physical barrier during centrifugation. Knowing which category an additive belongs to is central to understanding the whole tube system.

#195 · Tubes Colors and Additives

The chapter notes that two additives can both be called anticoagulants yet work completely differently. Give an example pair and explain the difference the exam tests.

Answer

Sodium citrate and heparin are both anticoagulants, but citrate chelates calcium away from the coagulation cascade while heparin potentiates antithrombin III without touching calcium at all. The exam tests whether a candidate understands this mechanistic difference rather than just being able to name both additives.

#196 · Tubes Colors and Additives

Why does the chapter emphasize understanding tube chemistry rather than memorizing a colour chart?

Answer

A technician who memorizes the colour chart without understanding why each additive exists is one mislabelled requisition or unusual order away from a serious error. Understanding the mechanism lets a candidate reason out the correct tube for an unfamiliar test order rather than simply recalling it from a table.

#197 · Tubes Colors and Additives

Between the light blue citrate tube and the lavender EDTA tube, which anticoagulant's chelation is reversible and which is irreversible?

Answer

Sodium citrate's calcium chelation in the light blue tube is reversible, allowing calcium to be added back for coagulation testing. EDTA's calcium chelation in the lavender tube is essentially irreversible under normal laboratory conditions, which is why EDTA blood cannot be used for coagulation testing.

#198 · Tubes Colors and Additives

Compare what a gold top SST and a green top PST have in common in terms of design.

Answer

Both use a separator gel with a density engineered to sit between cells and the liquid fraction after centrifugation. The SST pairs its gel with a silica clot activator to yield separated serum, while the PST pairs its gel with lithium heparin to yield separated plasma quickly.

#199 · Tubes Colors and Additives

Why can a specimen collected in a royal blue tube sometimes yield serum and other times yield plasma?

Answer

Royal blue tops are manufactured in more than one internal configuration: some are plain with no additive, producing serum after natural clotting like a trace-element-safe red top, while others contain EDTA or heparin, producing plasma. The requisition determines which configuration is required.

#200 · Tubes Colors and Additives

Why does the chapter describe the tube system as something to understand as principles rather than memorize tube by tube?

Answer

Several principles, such as fill volume calibration, mixing technique matched to additive type, and expiration dating, apply across nearly every tube rather than to just one. Understanding these shared principles lets a technician handle an unfamiliar tube correctly instead of relying on isolated, memorized facts.

#201 · Order of Draw and Equipment Selection

What phenomenon is the reason the order of draw exists?

Answer

Additive carryover, where a microscopic residue of the additive coating the inside of one tube remains on the needle and is carried into the next tube filled in the sequence. Even a trace amount can measurably shift the result of a sensitive assay. This is why tubes must be filled in a specific chemical-safe order rather than any convenient order.

#202 · Order of Draw and Equipment Selection

Which type of tube contains no additive at all, aside from a possible clot activator in some versions?

Answer

A plain serum tube, either a red-top with no additive or a gold-top serum separator tube with a clot activator and a gel barrier. Every other evacuated tube contains some anticoagulant, clot activator, or glycolysis inhibitor coated on its interior wall or suspended as liquid.

#203 · Order of Draw and Equipment Selection

How does EDTA anticoagulant prevent blood from clotting inside a lavender-top tube?

Answer

EDTA binds calcium ions, which are required for the coagulation cascade to proceed, so removing free calcium from the specimen prevents clot formation. Because EDTA is a potassium salt, the additive itself also introduces potassium into the tube along with its anticoagulant action.

#204 · Order of Draw and Equipment Selection

What happens to a chemistry tube's calcium and potassium results if a trace of EDTA carries over into it from a prior tube?

Answer

The calcium result reads falsely low because free calcium has been artificially bound and removed from the measurable pool, and the potassium result reads falsely high because potassium was introduced from the EDTA additive itself. A physician reading that panel has no way to know the numbers were distorted by tube sequence rather than by the patient's actual physiology.

#205 · Order of Draw and Equipment Selection

Why can a falsely elevated potassium result from EDTA carryover be clinically dangerous?

Answer

Depending on the clinical context, a falsely elevated potassium value could trigger urgent and unnecessary intervention, since elevated potassium is treated as a medical emergency. The error would look identical to a real physiological abnormality, so nothing on the report would alert the physician that the cause was tube sequence rather than the patient's actual blood chemistry.

#206 · Order of Draw and Equipment Selection

What internationally recognized body established the standard order of draw taught and tested in Canadian phlebotomy practice?

Answer

The Clinical and Laboratory Standards Institute, commonly abbreviated CLSI, which is the internationally recognized reference body for laboratory procedure standards. Canadian phlebotomy training and certification are built directly on the CLSI framework rather than a separate national standard.

#207 · Order of Draw and Equipment Selection

What is drawn first in the standard order of draw when ordered, and why does it take top priority?

Answer

Blood culture bottles or tubes are drawn first, when a culture has been ordered. Sterility is the single highest priority because the specimen's entire purpose is to detect organisms actually present in the bloodstream, so any microbial contamination, from skin flora or from carryover residue on the needle, can produce a false positive that leads to unnecessary antibiotic treatment or a misdiagnosed infection.

#208 · Order of Draw and Equipment Selection

What stopper colour identifies the sodium citrate tube, and what testing is it used for?

Answer

A light blue stopper identifies the sodium citrate tube, used for coagulation testing such as prothrombin time and activated partial thromboplastin time. It is drawn second in the standard sequence, immediately after any required blood culture.

#209 · Order of Draw and Equipment Selection

Give two reasons the sodium citrate tube is drawn second in the order of draw.

Answer

Coagulation assays are extraordinarily sensitive to the exact ratio of blood to anticoagulant, so carryover from a tube drawn earlier, especially one with a clot activator or a different anticoagulant, can meaningfully distort clotting time results. Drawing citrate early also minimizes how many prior tubes' additives could have contaminated the needle before reaching this highly sensitive specimen.

#210 · Order of Draw and Equipment Selection

If coagulation testing is the only test ordered, should the phlebotomist still observe the citrate tube's position in the standard sequence?

Answer

Yes, many laboratories still draw citrate in its standard second position even when it is the sole requested test. Standardizing the sequence for every draw reduces the chance of error across all the draws a technician performs in a shift, rather than relying on the phlebotomist to remember exceptions.

#211 · Order of Draw and Equipment Selection

Why are serum tubes positioned before the anticoagulated tubes that follow them in the order of draw?

Answer

Serum tubes contain a clot activator, and if that activator carried over into a subsequent anticoagulated tube, it could trigger unwanted clotting inside a tube that is meant to remain liquid and uncoagulated for testing. Drawing serum tubes before the anticoagulated tubes protects those downstream specimens from a contaminant that actively works against their intended function.

#212 · Order of Draw and Equipment Selection

What stopper colour identifies the heparin tube, and what does heparin do differently from citrate or EDTA?

Answer

A green stopper identifies the heparin tube. Heparin anticoagulates by potentiating antithrombin rather than by binding calcium, which is the mechanism citrate and EDTA both use. It is used for tests such as ammonia, certain stat chemistries, and some specialized panels needing anticoagulation without calcium chelation.

#213 · Order of Draw and Equipment Selection

Where does the heparin tube fall in the standard order of draw, and why?

Answer

Heparin is drawn fourth, after the serum tube and before the EDTA tube. It follows the serum tube for the same carryover logic that protects anticoagulated tubes from clot activator, and it precedes EDTA to limit the chance of heparin contaminating the hematology specimen drawn after it.

#214 · Order of Draw and Equipment Selection

What stopper colour identifies the EDTA tube, and what is it primarily used for?

Answer

A lavender stopper identifies the EDTA tube, the primary tube for complete blood counts and most hematology testing. It is drawn fifth in the standard sequence, after heparin and before the final glycolysis inhibitor tube.

#215 · Order of Draw and Equipment Selection

Why is the EDTA tube's position in the sequence described as a balance CLSI has established?

Answer

Drawing EDTA after heparin protects EDTA's own hematology results from contamination by additives drawn earlier. Drawing it before the final tube in the sequence protects that last tube from EDTA's own potent effects on calcium and potassium, so its position balances protecting itself against protecting the tube that comes after it.

#216 · Order of Draw and Equipment Selection

What stopper colour identifies the glycolysis inhibitor tube, and what does it contain?

Answer

A grey stopper identifies the glycolysis inhibitor tube, which typically contains sodium fluoride and often potassium oxalate. It is used chiefly for glucose and lactate testing and is drawn sixth and last in the standard order of draw.

#217 · Order of Draw and Equipment Selection

How does sodium fluoride preserve an accurate glucose result in the grey-top tube?

Answer

Sodium fluoride inhibits the enzyme enolase, which halts glycolysis and prevents red blood cells and other cellular elements from continuing to metabolize glucose after collection. Without fluoride, that ongoing metabolism would falsely lower the glucose result the longer the sample sat before analysis.

#218 · Order of Draw and Equipment Selection

Why is the grey-top fluoride tube drawn last in the standard order of draw?

Answer

Fluoride can interfere with a range of enzymatic assays if it carries into an earlier tube, so drawing it last avoids that risk entirely. Glucose and lactate testing is also comparatively tolerant of trace contamination from tubes drawn before it, making the final position the safest one for it to occupy.

#219 · Order of Draw and Equipment Selection

List the standard CLSI-based order of draw from first to last as taught in Canadian phlebotomy practice.

Answer

Blood culture bottles, sodium citrate light blue tube, serum red-top or gold-top tube, heparin green tube, EDTA lavender tube, and glycolysis inhibitor grey tube. Each position reflects a deliberate carryover-prevention rationale rather than an arbitrary sequence.

#220 · Order of Draw and Equipment Selection

When does a phlebotomist need to draw a discard tube before a citrate tube?

Answer

Only when a citrate tube is the very first tube collected through a winged, or butterfly, collection set. The discard tube allows the air-filled tubing of the winged set to be pulled empty before the citrate tube's own vacuum begins drawing blood, protecting its blood-to-anticoagulant ratio.

#221 · Order of Draw and Equipment Selection

Why does a winged, or butterfly, collection set create a special concern when citrate is drawn first?

Answer

A winged set connects the needle to the holder with a short length of flexible tubing that contains air before the first tube is attached and vacuum is applied. If citrate is drawn first, the vacuum must pull that air column through the tubing before blood begins to fill the tube, which can result in an underfilled citrate tube.

#222 · Order of Draw and Equipment Selection

What happens to coagulation test results if a citrate tube is underfilled because of air in a winged set's tubing?

Answer

The tube's vacuum is calibrated to draw a precise volume of blood to maintain the critical nine-to-one ratio of blood to citrate anticoagulant. An underfilled tube skews that ratio toward too much anticoagulant relative to blood volume, which falsely prolongs the coagulation test results.

#223 · Order of Draw and Equipment Selection

What kind of tube should be used as the discard tube before a first-drawn citrate tube on a winged set?

Answer

A small tube without an additive that would affect subsequent testing. Its purpose is only to let the vacuum pull the tubing's trapped air into it rather than into the citrate tube, after which the citrate tube can be attached and will fill to its correct volume.

#224 · Order of Draw and Equipment Selection

Is a discard tube ever required when a citrate tube is drawn first with a straight needle directly into a holder?

Answer

No. When a straight needle draws directly into a holder with no intervening tubing, there is no equivalent air space to displace, so no discard tube is required even when citrate is drawn first. The discard step is specific to the air-filled tubing segment of a winged set.

#225 · Order of Draw and Equipment Selection

In needle gauge numbering, does a larger gauge number mean a wider or a narrower needle bore?

Answer

A larger gauge number corresponds to a smaller, narrower needle bore, and a smaller gauge number corresponds to a larger, wider bore. This inverse relationship often confuses new students because it works opposite to ordinary intuition about numbers.

#226 · Order of Draw and Equipment Selection

What needle gauge is the standard choice for a typical adult patient with normal, easily palpable veins?

Answer

A 21 gauge needle is standard for a typical adult with normal, easily palpable veins. Its bore is wide enough to allow blood to flow at an appropriate rate into evacuated tubes without excessive turbulence.

#227 · Order of Draw and Equipment Selection

When is a 22 gauge needle generally selected instead of a 21 gauge needle?

Answer

A 22 gauge needle is generally selected when a patient's veins are smaller or more fragile than average. Its slightly narrower bore reduces the risk of collapsing a delicate vein under the pressure of the evacuated tube's vacuum draw.

#228 · Order of Draw and Equipment Selection

For which patients is a 23 gauge needle reserved, and what device is it most often part of?

Answer

A 23 gauge needle is reserved for pediatric patients, geriatric patients, and adults with particularly difficult veins in the hand or wrist. It is most often used within a winged infusion set rather than as a rigid straight needle, since the smallest practical bore reduces trauma to a vein that would not tolerate a wider needle.

#229 · Order of Draw and Equipment Selection

What can happen to a specimen if the needle gauge chosen is too narrow for the vacuum pressure of a standard evacuated tube?

Answer

A bore that is too narrow under strong vacuum pressure can damage red blood cells as they are forced through the small opening, a phenomenon known as hemolysis. Hemolysis renders many chemistry and hematology results unusable and requires the specimen to be redrawn.

#230 · Order of Draw and Equipment Selection

What risk does choosing too wide a needle bore for a small or fragile vein create?

Answer

A bore that is too wide for a small or fragile vein risks unnecessary patient discomfort, vein damage, and a failed or difficult collection. Gauge selection is therefore a balance between avoiding hemolysis from too narrow a bore and avoiding trauma from too wide a bore.

#231 · Order of Draw and Equipment Selection

Is needle gauge selection a fixed rule applied the same way to every patient?

Answer

No, matching gauge to vein condition is a clinical judgment made at the bedside for each individual patient, not a fixed rule applied identically across the board. The phlebotomist weighs vein size and fragility against the flow needs of the tests ordered.

#232 · Order of Draw and Equipment Selection

Describe the components of the evacuated tube system and identify when it is the default collection method.

Answer

The evacuated tube system consists of a straight needle threaded into a plastic holder that accepts tubes one after another. It remains the default collection method for a healthy adult with an accessible antecubital vein, meaning a vein at the bend of the elbow that is easily visualized or palpated and is not unusually fragile.

#233 · Order of Draw and Equipment Selection

Why is the evacuated tube system emphasized as the baseline collection approach in Canadian phlebotomy training?

Answer

It is efficient and allows multiple tubes to be filled from a single venipuncture, making it the technique most Canadian phlebotomy training emphasizes as the baseline approach before considering alternatives such as a winged set or a syringe.

#234 · Order of Draw and Equipment Selection

Describe the physical construction of a winged, or butterfly, infusion set.

Answer

A winged infusion set uses a shorter, thinner needle attached to flexible tubing with plastic wings that the phlebotomist holds to stabilize and guide the needle. It allows insertion at a shallower angle than a straight needle typically allows.

#235 · Order of Draw and Equipment Selection

List three situations in which a winged, or butterfly, set is indicated.

Answer

A winged set is indicated when the target vein is smaller or more fragile than average, when the only accessible vein is on the back of the hand rather than the antecubital area, and routinely for pediatric and geriatric patients whose veins tend to be smaller, more superficial, or less stable under a straight needle.

#236 · Order of Draw and Equipment Selection

Besides small, fragile, or pediatric and geriatric veins, in what other setting are butterfly sets frequently chosen?

Answer

Butterfly sets are frequently chosen for patients undergoing short-term draws in settings adjacent to intravenous therapy, where the smaller needle and flexible tubing offer greater control over needle placement compared with a rigid straight needle.

#237 · Order of Draw and Equipment Selection

When is the syringe method of blood collection indicated?

Answer

The syringe method is indicated for patients whose veins are especially difficult, small, or fragile, where a phlebotomist needs finer, hand-controlled pressure over the speed and force of the draw than an evacuated tube's fixed vacuum would allow.

#238 · Order of Draw and Equipment Selection

How does the syringe method physically collect blood, compared with the evacuated tube system?

Answer

In the syringe method, the phlebotomist manually draws back a plunger to fill a syringe barrel with blood, rather than relying on a tube's own internal vacuum as the evacuated tube system does. This gives the phlebotomist direct, hand-controlled command over draw speed and pressure.

#239 · Order of Draw and Equipment Selection

What device must be attached to a syringe after collection to fill laboratory tubes, and how does it work?

Answer

A blood transfer device must be attached to the syringe after collection, since a syringe alone cannot fill laboratory tubes directly. The device distributes the collected blood into the correct tubes, in the correct order of draw, by allowing each tube's own vacuum to pull the appropriate volume from the syringe through a sealed connector.

#240 · Order of Draw and Equipment Selection

Why is it never acceptable to remove the needle from a syringe and pour blood into tubes by hand?

Answer

Pouring blood by hand eliminates the closed, sterile transfer pathway, exposes the phlebotomist to needlestick and blood exposure risk, and makes it far more difficult to maintain accurate fill volumes and correct order of draw across multiple tubes. A blood transfer device must always be used instead.

#241 · Order of Draw and Equipment Selection

Are safety-engineered needle devices optional accessories in Canadian phlebotomy practice?

Answer

No, safety-engineered needle devices are the standard of practice rather than an optional accessory, driven by Canadian federal and provincial occupational health and safety requirements aimed at preventing needlestick injuries among healthcare workers.

#242 · Order of Draw and Equipment Selection

Name two mechanisms used by safety-engineered needle devices to protect against needlestick injury.

Answer

Retractable needles that withdraw into a protective housing after use, and needles fitted with a hinged or sliding shield that locks over the point immediately upon withdrawal from the vein. Both mechanisms cover or remove the sharp point before the phlebotomist could be accidentally stuck.

#243 · Order of Draw and Equipment Selection

Why have Canadian regulatory bodies mandated engineered sharps injury protection?

Answer

Needlestick injuries carry a genuine risk of bloodborne pathogen transmission, and regulatory bodies across Canadian jurisdictions have moved decisively toward mandating engineered sharps injury protection wherever a suitable safety device exists for the task.

#244 · Order of Draw and Equipment Selection

Is a needle holder used in the evacuated tube system safe to reuse on a second patient if a brand-new needle is attached?

Answer

No. The holder itself is designated for single use with a single patient, and this rule holds true even if a fresh, unused needle is attached to that same holder before use on a different patient. A new needle alone does not make the holder safe to reuse.

#245 · Order of Draw and Equipment Selection

Why can a needle holder become a cross-contamination risk even when it appears clean?

Answer

The interior of the holder can become contaminated with blood during use, even when no visible residue is apparent. Reusing a holder across patients, regardless of whether the needle attached to it is new, creates a cross-contamination risk that a fresh needle alone cannot eliminate.

#246 · Order of Draw and Equipment Selection

How should a needle holder be disposed of after a single patient's draw is complete?

Answer

Every holder is discarded into a sharps container after a single patient's draw is complete, along with the needle still attached to it. The holder and needle are treated as a single-use unit that is discarded together, not separated and reused.

#247 · Order of Draw and Equipment Selection

What is the physiological purpose of a tourniquet during venipuncture?

Answer

A tourniquet temporarily restricts venous return in the arm, causing veins to distend and become easier to palpate and access. This makes the vein a more reliable target for needle insertion before the draw begins.

#248 · Order of Draw and Equipment Selection

What are the two general categories of tourniquets available for venipuncture, and what material are disposable ones typically made from?

Answer

Tourniquets are available in single-patient-use disposable form and reusable multi-patient form. Disposable tourniquets are typically made of latex-free strip material and are discarded immediately after use on one patient.

#249 · Order of Draw and Equipment Selection

Why are disposable tourniquets increasingly preferred in Canadian healthcare settings?

Answer

Disposable tourniquets are increasingly preferred specifically because they eliminate any risk of transferring organic material or pathogens between patients through a shared strap, since each one is used on only a single patient before being discarded.

#250 · Order of Draw and Equipment Selection

What is required of a reusable tourniquet between every patient it is used on?

Answer

A reusable tourniquet requires thorough cleaning with an appropriate disinfectant between every patient according to the facility's infection control protocol. It cannot simply be wiped or reused untreated between patients.

#251 · Order of Draw and Equipment Selection

What should happen to a tourniquet that shows visible soiling, damage, or loss of elasticity?

Answer

Any tourniquet showing visible soiling, damage, or loss of elasticity must be removed from circulation entirely rather than cleaned and returned to use. Cleaning is not considered sufficient to restore a damaged or degraded tourniquet to service.

#252 · Order of Draw and Equipment Selection

Is the choice between disposable and reusable tourniquets left entirely to individual phlebotomist preference?

Answer

No, a phlebotomist choosing between disposable and reusable tourniquets is generally following that specific facility's own policy. The underlying principle that a tourniquet is a potential cross-contamination vector applies universally regardless of which type a facility uses.

#253 · Order of Draw and Equipment Selection

Why can a tube that looks perfectly filled, properly labelled, and correctly transported still produce a misleading laboratory result?

Answer

The mistake can happen at the moment of collection, inside the order the tubes were drawn in or the equipment chosen to draw them, rather than in any step visible after collection. A chemical or physical error at draw time leaves no outward sign on the finished specimen.

#254 · Order of Draw and Equipment Selection

What two categories of decisions govern every venipuncture before the needle touches the skin?

Answer

Which tubes will be filled and in what sequence, and which collection device is appropriate for the particular patient's veins and the particular clinical request. Both decisions look procedural but each rests on a chemical or physical principle that can silently corrupt a result if ignored.

#255 · Order of Draw and Equipment Selection

Why is drawing blood culture bottles before any other tube important even though the phlebotomist has already prepared the site?

Answer

Drawing the culture bottles first, before the needle has had contact with any other tube's additive or stopper, keeps the sample as clean as the venipuncture site preparation allows. Any tube drawn beforehand introduces an additional opportunity for contamination on the needle.

#256 · Order of Draw and Equipment Selection

What clinical consequence can result from a false positive blood culture caused by contamination?

Answer

A false positive blood culture can lead to unnecessary antibiotic treatment or a misdiagnosed infection. This is why sterility is treated as the single highest priority whenever a blood culture is part of the requisition.

#257 · Order of Draw and Equipment Selection

Which anticoagulant mechanism does heparin use that distinguishes it from citrate and EDTA?

Answer

Heparin works by potentiating antithrombin, whereas citrate and EDTA both work by binding calcium ions to prevent the coagulation cascade from proceeding. This different mechanism is part of why heparin occupies its own distinct position in the order of draw.

#258 · Order of Draw and Equipment Selection

What ratio must be maintained between blood and citrate anticoagulant for a coagulation specimen to be valid?

Answer

A nine-to-one ratio of blood to citrate anticoagulant must be maintained. The tube's vacuum is calibrated to draw a precise blood volume to achieve this ratio, and any underfill skews it toward too much anticoagulant relative to blood.

#259 · Order of Draw and Equipment Selection

Why can even a genuinely small amount of additive carryover matter for laboratory testing?

Answer

By volume, additive carryover is genuinely negligible, but many laboratory assays are sensitive enough that even a trace contaminant can measurably shift the result. Sensitivity of the assay, not the size of the contamination, is what makes carryover clinically significant.

#260 · Order of Draw and Equipment Selection

What single principle underlies both the order of draw and equipment selection as covered in this chapter?

Answer

Understanding the chemistry and physics behind each tube's additive and each device's mechanism well enough to predict, before the draw ever begins, exactly how a wrong choice would distort a result or endanger a patient or the phlebotomist. Both topics rest on this same predictive reasoning rather than rote sequence memorization.

#261 · Order of Draw and Equipment Selection

Does a gold-top serum separator tube contain any additive, and if so, what?

Answer

Yes, a gold-top serum separator tube contains a clot activator and a gel barrier, unlike a plain red-top tube which has no additive at all. Both are still classified together as serum tubes and occupy the third position in the order of draw.

#262 · Order of Draw and Equipment Selection

What is the clinical consequence if fluoride from a grey-top tube carries over into a tube drawn earlier in the sequence?

Answer

This scenario should not occur if the order of draw is followed correctly, since fluoride is drawn last specifically because it can interfere with a range of enzymatic assays if carried into an earlier tube. Its last position in the sequence is what prevents this interference from happening.

#263 · Order of Draw and Equipment Selection

Why is glucose and lactate testing considered a comparatively safe test to perform last in the order of draw?

Answer

Glucose and lactate testing is comparatively tolerant of trace contamination from tubes drawn before it, unlike coagulation or hematology testing, which is highly sensitive to carryover. This tolerance is part of why the fluoride tube for these tests occupies the final, least-protected position.

#264 · Order of Draw and Equipment Selection

What is an antecubital vein, and why does it matter for choosing a collection method?

Answer

An antecubital vein is a vein located at the bend of the elbow that is easily visualized or palpated and does not present unusual fragility. A healthy adult with an accessible antecubital vein is the typical candidate for the standard evacuated tube system rather than a winged set or syringe.

#265 · Order of Draw and Equipment Selection

Which needle gauge, straight or winged, is 23 gauge most commonly associated with in practice?

Answer

A 23 gauge needle is most often used within a winged infusion set rather than as a rigid straight needle. This pairing suits its typical patients, such as pediatric or geriatric patients and adults with difficult hand or wrist veins.

#266 · Order of Draw and Equipment Selection

What underlying physical principle must a phlebotomist balance when choosing a needle gauge?

Answer

The phlebotomist must balance hemolysis risk against vein trauma. Too narrow a bore under strong vacuum pressure can damage red blood cells and cause hemolysis, while too wide a bore in a small or fragile vein risks discomfort, vein damage, and a failed collection.

#267 · Order of Draw and Equipment Selection

Why does the accepted correction for a first-drawn citrate tube on a winged set use a discard tube rather than simply switching the draw order?

Answer

The chapter identifies the discard tube as the accepted correction specifically because it lets the air trapped in the winged set's tubing be pulled into a tube with no testing consequence, preserving citrate's required second position in the standard order of draw rather than reordering the sequence around it.

#268 · Order of Draw and Equipment Selection

What does hemolysis do to a blood specimen, and why does it require a repeat draw?

Answer

Hemolysis is damage to red blood cells forced through too narrow a needle bore under vacuum pressure, which ruptures the cells. This renders many chemistry and hematology results unusable, since the ruptured cells release their contents into the specimen and distort the measured values, requiring the sample to be redrawn.

#269 · Order of Draw and Equipment Selection

Why is standardizing equipment and sequence choices useful across an entire shift of draws, not just one difficult patient?

Answer

Standardization reduces the chance of error across every draw a technician performs in a shift, since a consistent habit is less likely to be forgotten under time pressure than an exception applied only in special cases. This is the same reasoning given for keeping citrate in its usual position even when it is the only test ordered.

#270 · Order of Draw and Equipment Selection

What is the relationship between the winged set's discard-tube rule and the straight-needle evacuated tube system?

Answer

The discard-tube rule applies only to a winged set because its flexible tubing holds a segment of air before the first tube is attached. A straight needle threaded directly into a holder has no equivalent air-filled space, so the same concern and the same correction do not apply to it.

#271 · Order of Draw and Equipment Selection

Why does the chapter describe order of draw and equipment selection as intertwined rather than separate topics?

Answer

Both look procedural on the surface but each rests on a chemical or physical principle that can silently corrupt a result if ignored, and choices in one area, such as using a winged set, directly affect the correct procedure in the other, such as whether a discard tube is needed before citrate.

#272 · Order of Draw and Equipment Selection

What must a phlebotomist be able to explain to demonstrate true mastery of this domain, according to the chapter's closing summary?

Answer

Why EDTA carryover falsely elevates potassium, why a discard tube matters only with a winged set drawing citrate first, why gauge selection balances hemolysis risk against vein trauma, and why a needle holder is never reused across patients even with a fresh needle attached. Reasoning through these facts, not just memorizing the sequence, is what this exam domain tests.

#273 · Order of Draw and Equipment Selection

Besides prothrombin time, what other coagulation test is the sodium citrate light blue tube used for?

Answer

Activated partial thromboplastin time, commonly abbreviated aPTT, is drawn in the sodium citrate light blue tube alongside prothrombin time. Both are coagulation assays that are extraordinarily sensitive to the exact blood-to-anticoagulant ratio in the tube.

#274 · Order of Draw and Equipment Selection

What two substances are typically combined in the grey-top glycolysis inhibitor tube?

Answer

Sodium fluoride and often potassium oxalate are typically combined in the grey-top tube. Sodium fluoride inhibits enolase to stop glycolysis, while potassium oxalate serves as the tube's anticoagulant component alongside the glycolysis inhibitor.

#275 · Order of Draw and Equipment Selection

Why does the syringe method give a phlebotomist an advantage over the evacuated tube system for a difficult vein?

Answer

The syringe method lets the phlebotomist manually control the plunger, giving finer, hand-controlled pressure over the speed and force of the draw than an evacuated tube's fixed vacuum allows. This adjustable control can reduce the risk of vein collapse in especially difficult, small, or fragile veins.

#276 · Patient ID and Patient Care

How many independent identifiers must be matched against the requisition before any specimen is collected?

Answer

At least two independent identifiers must be used and matched against the requisition or order before collection. A single identifier is never sufficient because names or birth dates can easily match more than one patient in a busy facility.

#277 · Patient ID and Patient Care

Name three identifiers that are acceptable for positive patient identification.

Answer

Acceptable identifiers include the patient's full legal name, date of birth, and in many Canadian facilities a unique health card number or medical record number assigned to that patient. Two of these must agree with the requisition and any specimen labels before collection proceeds.

#278 · Patient ID and Patient Care

Why is a room number or bed number never an acceptable patient identifier?

Answer

Patients are moved between rooms constantly and beds are reassigned within hours of a discharge, so a sign above a bed reflects whoever occupied that space when it was last updated, not necessarily the current occupant. Room or bed number is administrative information only and carries no evidentiary weight in confirming identity.

#279 · Patient ID and Patient Care

What is the correct way to phrase an identification question to a patient?

Answer

Ask an open question that forces the patient to supply the information themselves, such as, Can you tell me your full name and date of birth. This is preferred over reading the name aloud and asking for agreement, because open questions require the patient to generate the answer rather than simply confirm something stated to them.

#280 · Patient ID and Patient Care

Why is passive identification, asking a patient Are you John Smith, considered unsafe?

Answer

A patient who is confused, sedated, anxious, hearing impaired, cognitively affected by illness or medication, or simply eager to be agreeable may answer yes to a name that is not their own. Passive confirmation removes the safeguard that active, open-ended identification is designed to provide, so the CPT exam always treats the open question as the correct technique.

#281 · Patient ID and Patient Care

In the inpatient hospital setting, what physical object must be checked against the requisition?

Answer

The patient's wristband must be checked, not a whiteboard, a door placard, or a chart left at the foot of the bed. The phlebotomist compares the name, date of birth, and any medical record number on the wristband against the requisition and proceeds only once every element matches.

#282 · Patient ID and Patient Care

What should a phlebotomist do if an inpatient's wristband is missing, illegible, or damaged?

Answer

The draw must not proceed until a new band has been issued and applied by nursing staff. Identity must then be re-verified against the newly applied wristband before collection continues.

#283 · Patient ID and Patient Care

Is recognizing a patient from a previous visit an acceptable substitute for the documented identification process?

Answer

No. Familiarity is never a substitute for the documented identification process. Relying on memory has caused real specimen mix-ups when patients with similar appearances or names were housed in the same unit, so identification must always be verbally and physically confirmed.

#284 · Patient ID and Patient Care

How does patient identification differ in the outpatient or community collection setting compared to inpatient care?

Answer

There is no wristband, so identification relies on matching the requisition against government-issued photo identification where required, a provincial health card, or the patient stating their name and date of birth cross-referenced against a registration record. Some outpatient labs also ask for the patient's address as a third confirming detail when two patients share a name and birth date on the same day.

#285 · Patient ID and Patient Care

What should a phlebotomist do when a patient cannot participate in verbal identification, such as an unconscious or heavily sedated patient?

Answer

The correct response is never to proceed on a best guess. Facility protocols typically call for a second qualified staff member, such as the assigned nurse, to independently verify the patient's identity against the wristband and chart before the draw proceeds, with that verification documented.

#286 · Patient ID and Patient Care

What alternate identification methods might a facility use for patients who cannot speak for themselves?

Answer

Some facilities maintain an alternate identification protocol that may include photo verification stored in the electronic record or a designated identification bracelet system used specifically for patients who cannot speak for themselves. Whatever the local protocol, identification must be confirmed through an approved alternate method before collection, never assumed.

#287 · Patient ID and Patient Care

What minimum explanation must a phlebotomist give a patient before collection to satisfy informed consent?

Answer

The patient must be given a clear explanation of what is about to happen, including which tests have been ordered in general terms, what the collection will involve, and roughly what to expect in terms of sensation and duration. The patient must also understand that they have the right to refuse.

#288 · Patient ID and Patient Care

For a routine scheduled blood draw, is formal written consent required, and what is still expected of the phlebotomist?

Answer

Consent is generally treated as implied by the act of presenting for the appointment, but implied consent does not remove the phlebotomist's obligation to explain the procedure and seek the patient's explicit verbal assent before beginning. A simple statement such as, I'm going to draw a sample from your arm now, is that alright, satisfies this expectation.

#289 · Patient ID and Patient Care

Can a competent adult patient refuse a blood draw, and what happens if a phlebotomist proceeds anyway?

Answer

A patient's right to refuse is absolute; competent adult patients may decline for any reason or no stated reason at all. A phlebotomist who proceeds despite a refusal commits both an ethical violation and a form of assault in the legal sense.

#290 · Patient ID and Patient Care

What is the correct sequence of actions when a patient refuses a blood draw?

Answer

Stop the attempt, avoid pressuring or arguing the patient into compliance, document the refusal clearly and factually as the facility requires, and notify the ordering provider or nursing staff so the clinical team knows the requested testing did not occur.

#291 · Patient ID and Patient Care

What should documentation of a patient refusal include, and what should it avoid?

Answer

Documentation should record what was explained to the patient, that the patient declined, and the time of the refusal. It should avoid editorializing about the patient's reasoning or state of mind beyond what was directly observed.

#292 · Patient ID and Patient Care

What practical steps protect a patient's privacy during collection?

Answer

Drawing curtains or closing doors where available, keeping the exposed area limited to what collection requires, and avoiding conversations about the patient's condition or history within earshot of other patients or visitors all protect a patient's privacy during collection.

#293 · Patient ID and Patient Care

What does patient dignity require of a phlebotomist during an encounter?

Answer

Patients are entitled to be addressed respectfully, referred to by their correct name and preferred title, and not spoken about as though they were not present. This is a basic patient right that applies throughout the entire collection encounter.

#294 · Patient ID and Patient Care

What kinds of questions about a collection are within a phlebotomist's scope to answer?

Answer

Patients are entitled to have reasonable questions about the collection procedure answered clearly, such as what tube is being drawn, roughly how much blood, whether fasting was required, and how long results typically take to become available administratively.

#295 · Patient ID and Patient Care

If a patient asks what a high white count means or whether they are anemic, how should a phlebotomist respond?

Answer

The phlebotomist should tell the patient gently that the ordering provider is the appropriate person to discuss results and their meaning, and that the phlebotomist's role is limited to specimen collection. Interpreting results or speculating about a diagnosis falls outside the phlebotomist's scope of practice.

#296 · Patient ID and Patient Care

Why does the scope-of-practice boundary around interpreting results matter for both patient and phlebotomist?

Answer

It protects the patient from receiving incomplete or incorrect clinical information from someone not positioned to interpret it in the context of their full health picture, and it protects the phlebotomist from practising outside a defined scope.

#297 · Patient ID and Patient Care

What is the first step of therapeutic communication in a phlebotomy encounter?

Answer

A simple introduction, stating your name and role before doing anything else, orients an anxious or unfamiliar patient immediately and establishes the tone for the rest of the encounter.

#298 · Patient ID and Patient Care

When should a phlebotomist explain the collection procedure to the patient, relative to uncovering equipment?

Answer

A plain-language explanation of the procedure should be given before any equipment is uncovered or applied, so the patient is never surprised by a step they did not anticipate.

#299 · Patient ID and Patient Care

What does active listening mean in the context of a phlebotomy encounter, and why does it matter?

Answer

Active listening means actually attending to what the patient says, including hesitations, expressed fears, or questions about past difficult draws, rather than proceeding through a mental checklist while the patient talks. A patient who mentions a prior fainting episode or a difficult arm is giving genuinely useful clinical information that should change the approach taken.

#300 · Patient ID and Patient Care

Why should a phlebotomist avoid telling an anxious patient this won't hurt at all?

Answer

That statement is not true for the overwhelming majority of people, and when the patient inevitably feels the needle, the small dishonesty undermines the trust needed for the rest of the encounter and for future encounters with laboratory staff generally.

#301 · Patient ID and Patient Care

What is a more accurate and reassuring way to describe the sensation of needle insertion to a patient?

Answer

Explain honestly that the patient will feel a brief pinch or pressure as the needle is inserted, that the sensation is normal and expected, and that it typically passes quickly. Patients tend to tolerate an honestly described discomfort far better than an unexpected one.

#302 · Patient ID and Patient Care

How should communication with a pediatric patient differ from communication with an adult?

Answer

Explanations should be pitched to the child's developmental level rather than adult clinical language. A young child benefits from simple, concrete words and often from being told what will happen in terms of feeling and time rather than procedure names.

#303 · Patient ID and Patient Care

What role does a parent or caregiver play during a pediatric blood draw?

Answer

Involving a parent or caregiver is standard practice, both for comfort and for practical assistance holding a limb still. Distraction techniques such as a toy, a story, or the caregiver talking with the child during the draw meaningfully reduce distress and improve cooperation.

#304 · Patient ID and Patient Care

For very young infants, particularly neonates, which collection method is frequently preferred over venipuncture?

Answer

Capillary collection by heel stick is frequently preferred over venipuncture for very young infants, particularly neonates. The specific technique and site restrictions for this method are covered in the chapter on special populations and dermal puncture.

#305 · Patient ID and Patient Care

Why are geriatric patients at higher risk of bruising, tearing, or hematoma formation during venipuncture?

Answer

Aging skin loses collagen and becomes thinner and more fragile, which raises the risk of bruising, tearing, or hematoma formation. Veins in older patients are also often less elastic and more prone to rolling or collapsing under pressure.

#306 · Patient ID and Patient Care

What technique adjustments help offset the physical changes seen in geriatric patients?

Answer

Extra care in site selection, tourniquet tension, and needle angle helps offset the effects of thinner skin and less elastic veins. An unhurried, respectful pace also reduces the risk of a failed or traumatic draw.

#307 · Patient ID and Patient Care

What communication adjustments benefit many older patients during a blood draw?

Answer

Many older patients have hearing changes or slower processing of spoken instructions, so speaking clearly, facing the patient while speaking, and allowing extra time for questions and for the patient to settle into position all improve comfort and safety.

#308 · Patient ID and Patient Care

How should a phlebotomist adjust their communication approach for a patient with dementia or cognitive impairment?

Answer

Use simple, calm, and often repeated explanations, since a single explanation may not be retained from one moment to the next. Short sentences, a warm and unhurried tone, and consistency in what is said each time help these patients feel secure rather than confused.

#309 · Patient ID and Patient Care

Who should be involved when collecting from a patient with dementia who cannot provide informed consent themselves?

Answer

A caregiver, family member, or substitute decision-maker should be involved where one is present and where the situation allows, both to support the patient emotionally and to participate in the consent process in the manner the facility's policy and applicable provincial legislation require.

#310 · Patient ID and Patient Care

What is the preferred approach to communicating with a patient who has a significant language barrier?

Answer

Wherever possible, a qualified medical interpreter, present in person or accessed by phone or video interpretation service, should be used rather than a family member acting as an informal translator.

#311 · Patient ID and Patient Care

Why is using a family member as an informal interpreter for a medical explanation problematic?

Answer

A family member may unintentionally simplify, omit, or editorialize what is being explained, and using a minor as an interpreter for medical information is generally inappropriate. A qualified interpreter provides an accurate, complete rendering of the explanation and the patient's responses.

#312 · Patient ID and Patient Care

Why does interpreter accuracy matter especially when the topic includes consent?

Answer

The topic of a patient's right to refuse a procedure requires precise communication, and an inaccurate or incomplete informal translation could cause a patient to consent or refuse without truly understanding what was explained. A qualified interpreter protects the integrity of the consent process.

#313 · Patient ID and Patient Care

What does cultural sensitivity look like in everyday phlebotomy practice?

Answer

Respecting personal space, modesty, and any expressed preference around touch or the gender of the phlebotomist performing the draw, where such accommodation is reasonably possible without compromising patient safety or unduly delaying urgent testing, reflects both professionalism and basic respect.

#314 · Patient ID and Patient Care

What should a phlebotomist do when a patient's cultural or personal accommodation request genuinely cannot be met?

Answer

For instance in an emergency where only one qualified phlebotomist is available, the situation should be explained to the patient honestly and kindly rather than ignored or dismissed.

#315 · Patient ID and Patient Care

What is the professional response when a patient becomes upset, agitated, or uncooperative during a draw?

Answer

Remain calm, lower the emotional temperature of the interaction rather than match it, and avoid arguing or attempting to physically compel cooperation. This applies whether the behaviour stems from pain, a prior traumatic healthcare experience, or an underlying condition affecting mood.

#316 · Patient ID and Patient Care

If a patient cannot be calmed and safely collected from, what should the phlebotomist do?

Answer

The appropriate step is to pause the attempt, offer to reschedule, and involve nursing staff, a supervisor, or security personnel if the situation involves any risk to the patient's safety or the phlebotomist's own safety.

#317 · Patient ID and Patient Care

Is it ever appropriate to use force or restraint to complete a routine blood draw on an uncooperative patient?

Answer

No. Proceeding by force or restraint is never appropriate for a routine blood draw and exposes both the patient and the facility to serious harm and liability.

#318 · Patient ID and Patient Care

What should be documented after a difficult or uncooperative patient encounter, whether the draw succeeded or not?

Answer

Every such encounter should be documented factually: what occurred, what was explained, what the outcome was, and who else was involved or notified.

#319 · Patient ID and Patient Care

What is the purpose of chain of custody procedures for forensic specimens?

Answer

Chain of custody procedures ensure that a specimen can be traced, without any gap or possibility of tampering, from the moment of collection through every point of handling to final laboratory analysis and reporting, because the result may be used as evidence in a legal or disciplinary proceeding.

#320 · Patient ID and Patient Care

For which types of collections is a heightened chain of custody process typically required?

Answer

Chain of custody procedures most commonly apply to certain drug and alcohol testing collections performed for employment, legal, or regulatory purposes, where the result may later be used as evidence.

#321 · Patient ID and Patient Care

What form of identification does a forensic chain of custody collection typically require, compared to routine clinical practice?

Answer

A forensic collection typically requires government-issued photo identification, rather than the verbal or wristband identification that is acceptable in routine clinical practice.

#322 · Patient ID and Patient Care

What role does a witness play in a chain of custody collection?

Answer

A witness must be present for the collection and their signature is recorded on the chain of custody form, as part of the heightened documentation required to make the specimen legally defensible.

#323 · Patient ID and Patient Care

What must happen to a forensic specimen container immediately after collection, and in whose presence?

Answer

The specimen container must undergo tamper-evident sealing in the presence of the donor, and the donor must sign to confirm that the sealed specimen is theirs.

#324 · Patient ID and Patient Care

What must be logged every time custody of a forensic specimen changes hands?

Answer

Every transfer of custody, whether to a courier, a testing laboratory, or a storage facility, must be logged with a date, time, and signature to maintain an unbroken chain of custody.

#325 · Patient ID and Patient Care

What happens if a phlebotomist conflates routine clinical identification with forensic chain of custody, or skips a step because it feels redundant?

Answer

Compromising or omitting any step of the forensic process, even because it feels redundant with the clinical identification already performed, compromises the legal defensibility of the result regardless of how accurate the laboratory analysis itself turns out to be.

#326 · Patient ID and Patient Care

Why can patient identification errors have such serious consequences?

Answer

A misidentified specimen can lead to a transfusion reaction, a missed diagnosis, an unnecessary treatment, or a delayed one. Identification errors are among the most preventable and most consequential mistakes a phlebotomist can make.

#327 · Patient ID and Patient Care

Why is a first name, last name, or date of birth alone never sufficient for patient identification?

Answer

A single identifier can easily match more than one patient in a busy hospital or clinic, particularly among patients who share a common surname or who were born on the same day. Two independent identifiers must always agree with the requisition.

#328 · Patient ID and Patient Care

In a rare case where two scheduled outpatients share an identical name and birth date, what additional detail can help resolve the ambiguity?

Answer

Many outpatient labs ask the patient to state their address as a third confirming detail in this situation, rather than proceeding on assumption. The phlebotomist should be prepared to resolve the ambiguity rather than guess.

#329 · Patient ID and Patient Care

What must be true of every element on a wristband before an inpatient draw proceeds?

Answer

The name, date of birth, and any medical record number printed on the wristband must all match the requisition. The phlebotomist proceeds only once every element matches, not just some of them.

#330 · Patient ID and Patient Care

Give an example of a clinical detail a patient might mention that should change a phlebotomist's approach to the draw.

Answer

A patient who mentions they have fainted during a previous draw, or that a particular arm is difficult to access, is giving genuinely useful clinical information. Active listening means this detail should influence the site chosen or the precautions taken, not be ignored.

#331 · Patient ID and Patient Care

What is the difference between implied consent and the explicit verbal assent a phlebotomist should still seek?

Answer

Implied consent is treated as granted by the act of a patient presenting for an appointment, but it does not remove the obligation to explain the procedure and seek explicit verbal assent, such as asking the patient directly if it is alright to proceed, before beginning.

#332 · Patient ID and Patient Care

What is the correct tone and approach when explaining a delayed or denied cultural accommodation to a patient?

Answer

The situation should be explained to the patient honestly and kindly, rather than ignored or dismissed, particularly when only one qualified phlebotomist is available and a requested accommodation genuinely cannot be met right away.

#333 · Patient ID and Patient Care

How should a phlebotomist introduce themselves at the start of a patient encounter?

Answer

The encounter should begin with a simple introduction stating the phlebotomist's name and role before doing anything else. This orients an anxious or unfamiliar patient immediately and sets a professional tone.

#334 · Patient ID and Patient Care

Why is it inappropriate for a phlebotomist to speak about a patient's condition within earshot of other patients or visitors?

Answer

Doing so violates the patient's right to privacy during collection. Privacy in practice includes limiting exposure, using curtains or closed doors where available, and keeping any discussion of the patient's condition or history out of earshot of others.

#335 · Patient ID and Patient Care

What underlying rule governs identification of a patient who cannot participate verbally, regardless of the specific local protocol used?

Answer

The underlying rule is constant: identification must be confirmed through an approved alternate method before collection, never assumed because the patient appears to match the chart or because time pressure makes verification feel unnecessary.

#336 · Patient ID and Patient Care

What distinguishes the identification standard required for a routine clinical draw from that required for a legally significant forensic draw?

Answer

Routine clinical identification accepts verbal confirmation of two identifiers or a checked wristband, while a forensic chain of custody collection requires government-issued photo identification, a witness, tamper-evident sealing, and the donor's signature, because the result may later serve as legal evidence.

#337 · Patient ID and Patient Care

Why should a phlebotomist avoid arguing with or pressuring a patient who has stated a refusal?

Answer

The patient's right to refuse is absolute, so arguing or pressuring the patient into compliance is inappropriate. The correct response is to stop, document the refusal factually, and notify the ordering provider or nursing staff.

#338 · Patient ID and Patient Care

How should a phlebotomist balance throughput pressure against the extra time an elderly patient may need?

Answer

None of the extra care an elderly patient needs should be rushed for the sake of throughput. An unhurried, respectful pace with an elderly patient reduces the risk of a failed or traumatic draw far more than it costs in time.

#339 · Patient ID and Patient Care

What is a testable competency this chapter identifies regarding chain of custody collections specifically?

Answer

Recognizing which collections require the heightened chain of custody process, and following it precisely rather than substituting familiar clinical habits, is described as a distinct and testable competency within the patient identification and patient care domain.

#340 · Patient ID and Patient Care

Why does the CPT exam place particular weight on the patient identification domain?

Answer

Because identification errors are among the most preventable and most consequential mistakes a phlebotomist can make in real practice, with the potential to cause a transfusion reaction, a missed diagnosis, or a delayed or unnecessary treatment, the CPT exam treats this domain with the weight it deserves.

#341 · Anatomy and Physiology

What is the primary function of the heart within the circulatory system?

Answer

The heart acts as a single central pump that pushes blood through a closed network of vessels to every tissue in the body. Its forceful, rhythmic contractions generate the pressure needed to move blood outward through the arteries and, ultimately, to return it through the veins.

#342 · Anatomy and Physiology

What distinguishes arteries from veins in terms of pressure and wall structure?

Answer

Arteries carry oxygenated blood away from the heart under relatively high pressure and have thick, muscular walls to withstand this force, which is also why a pulse can be felt over them. Veins carry deoxygenated blood back toward the heart under much lower pressure and have thinner walls.

#343 · Anatomy and Physiology

Why do veins rely on one-way valves?

Answer

Because the pressure driving blood through veins is comparatively weak, especially in the limbs where blood must travel against gravity, veins depend on one-way valves positioned along their length. These valves permit blood to flow only toward the heart and work with contracting skeletal muscle to prevent backflow.

#344 · Anatomy and Physiology

What happens at the level of the capillaries?

Answer

Capillaries are microscopic vessels with walls only a single cell thick, positioned between the arterial and venous systems. It is at this level that the actual exchange of oxygen, carbon dioxide, nutrients, and waste products takes place between the blood and the surrounding tissue.

#345 · Anatomy and Physiology

Give three reasons why phlebotomy is built around venous rather than arterial blood collection.

Answer

Veins near the skin surface, particularly in the antecubital area, are relatively accessible without specialized equipment. The low pressure inside a vein makes puncture comparatively safe, since bleeding is controlled quickly with pressure after the draw. Venous blood, having circulated and mixed thoroughly, also provides a representative whole-body sample for most laboratory tests.

#346 · Anatomy and Physiology

Why is arterial puncture avoided by the general phlebotomy technician?

Answer

The high pressure inside an artery means a puncture carries real risk of uncontrolled bleeding, hematoma formation, and arterial spasm. Controlling bleeding afterward requires sustained direct pressure for much longer than a venous draw, so arterial blood gas sampling is reserved for staff with additional specialized training.

#347 · Anatomy and Physiology

When is capillary blood collection deliberately chosen instead of venipuncture?

Answer

Capillary blood, obtained by a skin puncture rather than inserting a needle into a vessel, is used when only a very small volume of blood is required. Typical situations include newborn screening programs and certain point-of-care glucose testing scenarios.

#348 · Anatomy and Physiology

What is the antecubital fossa and why does it matter to phlebotomy?

Answer

The antecubital fossa is the shallow depression on the inner surface of the elbow. It is the primary collection site for the great majority of venipunctures, which is why a technician must know its venous anatomy in detail.

#349 · Anatomy and Physiology

Why is the median cubital vein generally the preferred venipuncture site?

Answer

It tends to be large in diameter and well anchored to surrounding tissue, so it rolls less readily under the needle, and its position makes venipuncture relatively less painful. It also sits at a comfortable distance from major nerves and the brachial artery, reducing the risk of inadvertent injury.

#350 · Anatomy and Physiology

Where does the cephalic vein run, and when is it typically used?

Answer

The cephalic vein runs along the lateral, or thumb, side of the forearm and extends up through the antecubital area. It becomes a useful alternative when the median cubital vein is unavailable due to scarring, bruising, or being poorly developed in that patient.

#351 · Anatomy and Physiology

What technical challenge can the cephalic vein present compared to the median cubital vein?

Answer

The cephalic vein can sit slightly deeper beneath the skin in some patients and has a greater tendency to roll away from the needle during insertion. This means the technician may need to anchor the skin more firmly before attempting the draw.

#352 · Anatomy and Physiology

Where is the basilic vein located, and when should it be used?

Answer

The basilic vein lies along the medial side of the arm, closer to the body's midline. It is generally used only as a site of last resort, when neither the median cubital nor the cephalic vein presents a viable option.

#353 · Anatomy and Physiology

Why is the basilic vein considered higher risk than the median cubital or cephalic veins?

Answer

The basilic vein's position places it in closer proximity to the brachial artery and the median nerve. This proximity is the reason it is treated as a site of last resort rather than a routine first choice for venipuncture.

#354 · Anatomy and Physiology

What is the H-pattern of antecubital venous arrangement?

Answer

In the H-pattern, the median cubital vein forms a relatively straight, horizontal connecting segment between the cephalic vein on the lateral side and the basilic vein on the medial side. Together the three veins produce an arrangement that visually resembles the letter H.

#355 · Anatomy and Physiology

What is the M-pattern of antecubital venous arrangement?

Answer

In the M-pattern, an accessory median vein branches off and creates an additional connecting pathway, so the median cephalic and median basilic segments together with the accessory vein form a shape closer to the letter M. It is a normal anatomical variation seen with meaningful frequency in the population.

#356 · Anatomy and Physiology

Is the H-pattern more normal than the M-pattern of antecubital veins?

Answer

No. Neither pattern is more normal than the other; both occur with meaningful frequency in the general population. A competent technician assesses the actual vessels present in front of them on each patient rather than assuming a textbook H-pattern will always be found.

#357 · Anatomy and Physiology

What two anatomical structures lie in close proximity to the basilic vein?

Answer

The brachial artery and the median nerve both run in close anatomical proximity to the basilic vein, deeper in the tissue but near enough that a poorly controlled needle trajectory can reach them. This is the physiological basis for the vein's elevated risk status.

#358 · Anatomy and Physiology

What is probing during venipuncture, and why is it unsafe?

Answer

Probing is redirecting the needle beneath the skin while searching for a vein after a failed first attempt. In the antecubital fossa generally, and over the basilic vein specifically, this blind redirection risks lacerating the brachial artery or striking the median nerve.

#359 · Anatomy and Physiology

What can happen if the brachial artery is lacerated during a failed venipuncture attempt?

Answer

Lacerating the brachial artery can cause significant bleeding and hematoma formation. This is one of the key reasons blind needle redirection near the basilic vein is considered unsafe practice.

#360 · Anatomy and Physiology

What can happen if the median nerve is struck during venipuncture?

Answer

Striking the median nerve can cause sharp, radiating pain and, in more serious cases, lasting nerve damage affecting sensation or function in the hand and forearm. This is one of the two major hazards associated with probing near the basilic vein.

#361 · Anatomy and Physiology

What is the correct response when a venipuncture attempt does not obtain blood flow?

Answer

The correct response is to withdraw the needle fully, reassess the patient's anatomy and the planned approach, and either attempt a different site or a different vein. This withdraw-and-reassess principle carries the highest stakes when the basilic vein is involved.

#362 · Anatomy and Physiology

What are the two main components of whole blood?

Answer

Whole blood is composed of a liquid portion, called plasma, and a cellular portion containing red blood cells, white blood cells, and platelets. Plasma makes up slightly more than half of blood volume in a healthy adult.

#363 · Anatomy and Physiology

What substances are dissolved in plasma?

Answer

Plasma consists mostly of water along with proteins such as albumin, clotting factors including fibrinogen, electrolytes such as sodium and potassium, hormones, nutrients, and waste products awaiting elimination. It is the liquid portion of whole blood.

#364 · Anatomy and Physiology

What is the defining chemical difference between plasma and serum?

Answer

The presence or absence of fibrinogen and other consumed clotting factors is the defining chemical difference. Plasma still contains fibrinogen intact, whereas serum is essentially depleted of fibrinogen because the clotting process consumes it while forming the clot.

#365 · Anatomy and Physiology

How is serum obtained from a blood sample?

Answer

Serum is what remains after blood has been allowed to clot and the clot has been removed, typically by centrifugation. A tube with no anticoagulant, such as a plain red-top tube, or a serum separator tube allows the blood to clot naturally before this process.

#366 · Anatomy and Physiology

How is plasma obtained from a blood sample, and what tube additives make this possible?

Answer

A tube containing an anticoagulant additive, such as EDTA or heparin, prevents the blood inside it from clotting at all. When this tube is centrifuged, the cells are separated from the still-intact liquid portion, and the result is plasma rather than serum.

#367 · Anatomy and Physiology

Which blood cells are the most numerous, and what is their function?

Answer

Red blood cells, also called erythrocytes, are by far the most numerous cell type in blood and are responsible for carrying oxygen throughout the body. Their oxygen-carrying capacity depends on hemoglobin, an iron-containing protein packed inside each cell.

#368 · Anatomy and Physiology

What is hemoglobin, and what does it do?

Answer

Hemoglobin is an iron-containing protein packed inside each red blood cell. It binds oxygen in the lungs and releases it in the tissues, giving red blood cells their oxygen-carrying capacity.

#369 · Anatomy and Physiology

What is the role of white blood cells, and what is a complete blood count expected to report about them?

Answer

White blood cells, or leukocytes, serve as the body's immune defence, identifying and responding to infection, foreign material, and abnormal cells. A complete blood count typically reports both the total white cell count and a breakdown by the several distinct types that exist.

#370 · Anatomy and Physiology

What are platelets, and what is their central role?

Answer

Platelets, also called thrombocytes, are not true cells but rather small cell fragments. They play a central role in stopping bleeding when a blood vessel is damaged.

#371 · Anatomy and Physiology

What is hemostasis?

Answer

Hemostasis is the physiological process by which the body limits blood loss after vessel injury. It unfolds in two broadly recognized stages, primary hemostasis and secondary hemostasis.

#372 · Anatomy and Physiology

What happens during primary hemostasis?

Answer

Primary hemostasis begins almost immediately after a vessel wall is breached, when platelets are activated, adhere to the site of injury, and aggregate together to form a temporary platelet plug. This plug provides a fast but relatively fragile initial seal.

#373 · Anatomy and Physiology

What happens during secondary hemostasis?

Answer

Secondary hemostasis follows and reinforces the platelet plug through the coagulation cascade, a sequence of clotting factors activating one another in order, ultimately converging on the conversion of fibrinogen into fibrin. Fibrin forms a mesh that stabilizes the plug into a durable clot.

#374 · Anatomy and Physiology

What is fibrin, and what does it do?

Answer

Fibrin is formed by the conversion of fibrinogen at the end of the coagulation cascade. It forms a mesh of strands that stabilizes and strengthens the temporary platelet plug into a durable clot during secondary hemostasis.

#375 · Anatomy and Physiology

Why is a citrate tube used for coagulation testing?

Answer

The coagulation cascade depends on calcium at several key steps. Citrate is a calcium-chelating additive that binds available calcium, interrupting the cascade before it can run to completion, which keeps the sample in a state suitable for accurate coagulation studies rather than allowing it to clot inside the tube.

#376 · Anatomy and Physiology

What are the two principal layers of skin relevant to phlebotomy?

Answer

The epidermis is the thin outermost layer that provides a protective barrier. Beneath it lies the dermis, a thicker layer containing blood vessels, nerve endings, and the capillary beds that skin puncture technique is designed to access.

#377 · Anatomy and Physiology

What lies beneath the dermis, and why does this matter for capillary puncture?

Answer

Beneath the dermis lies subcutaneous tissue, and beneath that, particularly in the extremities of infants, bone can be located surprisingly close to the skin surface. This anatomical reality is the basis for strict depth limits on capillary puncture devices.

#378 · Anatomy and Physiology

What risk does a capillary puncture that is too deep carry, and where is this especially significant?

Answer

A puncture that penetrates too deeply risks striking bone, an outcome that carries a real risk of infection and injury. This concern is especially significant in infant heel stick procedures, where the distance from skin to the underlying calcaneus bone is far shorter than in an adult limb.

#379 · Anatomy and Physiology

What is the lymphatic system, and how does it relate to blood circulation?

Answer

The lymphatic system is a network of vessels and nodes that collects excess fluid from tissues and returns it to the bloodstream, while also playing a central role in immune function. It operates separately from the blood circulatory system but has direct implications for phlebotomy site selection.

#380 · Anatomy and Physiology

What is lymphedema, and how can it arise?

Answer

Lymphedema is chronic swelling caused by fluid accumulating in an affected limb because it can no longer drain normally. It can result when lymph nodes are surgically removed, as commonly occurs during treatment for breast cancer involving a mastectomy.

#381 · Anatomy and Physiology

Why does venipuncture on a limb affected by lymphedema carry elevated risk?

Answer

Local circulation and fluid balance are already compromised in a limb with lymphedema, healing may be impaired, and the risk of infection is increased because the limb's normal immune drainage pathway has been altered.

#382 · Anatomy and Physiology

What is a dialysis fistula or graft, and why should a limb with one be avoided for other venipunctures?

Answer

A dialysis fistula or graft is a surgically created vascular access point used for hemodialysis. Puncturing elsewhere on the same limb risks damaging a structure that is often medically essential to the patient and disrupting the specialized blood flow pattern the fistula depends on.

#383 · Anatomy and Physiology

Why is drawing blood from or near an active IV line site a concern?

Answer

Blood drawn from or near an active IV site can be diluted or otherwise altered by the infused fluid, and the vessel itself may already be compromised at that location. This is a related but distinct concern from lymphedema and fistula sites.

#384 · Anatomy and Physiology

What single underlying principle explains why lymphedema, fistula, and IV sites are all avoided for venipuncture?

Answer

In each case, the underlying physiological reasoning, whether lymphedema risk, infection risk, or disruption of specialized or already-compromised blood flow, is what justifies avoiding the site. These are not arbitrary rules but consequences of the actual anatomy and physiology involved.

#385 · Anatomy and Physiology

Why does venous blood provide a more representative whole-body sample than capillary blood for most laboratory tests?

Answer

Venous blood has circulated through the body and mixed thoroughly before it reaches the collection site, giving it a composition representative of the whole body. This is one of the converging reasons phlebotomy relies on venous rather than capillary sampling for most standard laboratory tests.

#386 · Anatomy and Physiology

How does understanding of blood pressure differences between vessel types guide the choice of puncture site?

Answer

Arteries carry blood under high pressure, making them risky to puncture, while veins carry blood under low pressure, making venous bleeding quick and predictable to control. This pressure difference is a core reason phlebotomy technique targets veins rather than arteries for routine collection.

#387 · Anatomy and Physiology

What role does skeletal muscle contraction play in venous blood flow?

Answer

Because venous pressure is comparatively weak, especially in the limbs where blood travels against gravity, the contraction of surrounding skeletal muscle works together with the one-way valves in veins to help keep blood moving forward toward the heart.

#388 · Anatomy and Physiology

Why is arterial blood gas sampling considered outside the scope of routine phlebotomy practice?

Answer

Arterial blood gas sampling measures oxygen, carbon dioxide, and pH directly from arterial blood and is a distinct, specialized skill reserved for staff who have received additional training specific to that procedure, given the elevated risks of puncturing a high-pressure vessel.

#389 · Anatomy and Physiology

Why does a phlebotomy technician not need to memorize every clotting factor in the coagulation cascade?

Answer

A technician only needs a practical understanding that the cascade depends on calcium at several key steps, which explains the purpose of citrate additive tubes. The full clinical detail of every individual factor is not required for safe, competent collection practice.

#390 · Anatomy and Physiology

What overall principle connects vein selection, tube additive choice, and site avoidance in phlebotomy practice?

Answer

Each of these practical decisions follows directly from an underlying anatomical or physiological reason, such as vessel pressure, proximity to nerves and arteries, clotting chemistry, or compromised circulation, rather than being an arbitrary rule to memorize without context.

#391 · Medical Terminologies and Abbreviations

What is a root word in medical terminology and what does it carry?

Answer

A root word carries the core meaning of a medical term, usually naming a body structure, substance, or process. Roots rarely stand alone; they combine with prefixes and suffixes to build the specific words used on requisitions and in laboratory reports.

#392 · Medical Terminologies and Abbreviations

What do the roots hem/o and hemat/o mean?

Answer

Both hem/o and hemat/o mean blood. Either form may appear depending on what letter or piece follows it, as in hemolysis or hematology.

#393 · Medical Terminologies and Abbreviations

What does the root cyt/o mean, and give an example term.

Answer

Cyt/o means cell. It appears in thrombocytopenia, where cyt/o refers to platelets (sometimes called thrombocytes), combined with thromb/o for clot and the suffix -penia for deficiency.

#394 · Medical Terminologies and Abbreviations

What do the roots ven/o, arteri/o, and angi/o refer to?

Answer

Ven/o refers to a vein, arteri/o refers to an artery, and angi/o means vessel in general without specifying vein or artery. These roots distinguish venous, arterial, and generic vessel terminology.

#395 · Medical Terminologies and Abbreviations

What do the roots erythr/o, leuk/o, and thromb/o mean?

Answer

Erythr/o means red, leuk/o means white, and thromb/o refers to a clot. These roots combine with other pieces to describe red blood cells, white blood cells, and clotting-related conditions.

#396 · Medical Terminologies and Abbreviations

What do the prefixes hypo- and hyper- mean?

Answer

Hypo- means below or under, as in a blood glucose result lower than normal. Hyper- means above or excessive, the opposite condition, as in hyperglycemia.

#397 · Medical Terminologies and Abbreviations

What do the prefixes brady- and tachy- mean?

Answer

Brady- means slow, most familiar from bradycardia, an abnormally slow heart rate. Tachy- means fast, its opposite, as in tachycardia.

#398 · Medical Terminologies and Abbreviations

What do the prefixes peri-, sub-, and intra- mean?

Answer

Peri- means around or surrounding, sub- means beneath or under, and intra- means within. These prefixes describe position or location relative to a body structure.

#399 · Medical Terminologies and Abbreviations

What does the suffix -emia mean, and give an example.

Answer

The suffix -emia refers to a blood condition, so any word ending in -emia describes something present in, or missing from, the bloodstream. Hyperglycemia uses this suffix to describe excessive sugar in the blood.

#400 · Medical Terminologies and Abbreviations

What do the suffixes -itis and -osis mean?

Answer

The suffix -itis means inflammation, as in phlebitis, inflammation of a vein. The suffix -osis means an abnormal condition or an increase, and is often used more neutrally than -itis.

#401 · Medical Terminologies and Abbreviations

What does the suffix -penia mean?

Answer

The suffix -penia means a deficiency or an abnormally low count of something. In thrombocytopenia it describes a deficiency of clotting cells, the clinical term for a low platelet count.

#402 · Medical Terminologies and Abbreviations

What does the suffix -lysis mean, and how does it apply to hemolysis?

Answer

The suffix -lysis means breakdown or destruction. In hemolysis, the root hem/o for blood combines with -lysis to give a literal meaning of breakdown of blood, matching the rupture of red blood cells that releases hemoglobin into surrounding fluid.

#403 · Medical Terminologies and Abbreviations

What does the suffix -stasis mean, and how does it relate to phlebotomy?

Answer

The suffix -stasis means stopping or controlling. Hemostasis combines hem/o for blood with -stasis, describing the process, and the practical goal after venipuncture, of stopping bleeding through direct pressure until clotting seals the puncture.

#404 · Medical Terminologies and Abbreviations

What is the difference between -graphy and -gram as suffixes?

Answer

The suffix -graphy refers to the process of recording an image, while -gram refers to the resulting record itself. One describes the act of imaging, the other the finished output of that process.

#405 · Medical Terminologies and Abbreviations

What do the suffixes -ectomy and -pathy mean?

Answer

The suffix -ectomy means surgical removal. The suffix -pathy means disease. Both attach to a root to describe a procedure or condition affecting that structure.

#406 · Medical Terminologies and Abbreviations

Break down the word arteriosclerosis into its parts and give its literal meaning.

Answer

Arteriosclerosis combines arteri/o, meaning artery, with -sclerosis, a suffix meaning hardening. Its literal meaning is hardening of the arteries.

#407 · Medical Terminologies and Abbreviations

Break down the word venipuncture into its parts.

Answer

Venipuncture combines ven/i, meaning vein, with puncture. It is a straightforward description of puncturing a vein to obtain a blood sample, the phlebotomist's core procedure.

#408 · Medical Terminologies and Abbreviations

What do the directional terms proximal and distal describe?

Answer

Proximal and distal describe distance from the point where a limb attaches to the trunk. Proximal means closer to that attachment point, while distal means farther away from it.

#409 · Medical Terminologies and Abbreviations

Why might a phlebotomist document a new draw site as distal to a previous IV site?

Answer

Selecting a site distal to, or farther down the arm from, an existing IV avoids drawing blood that has been diluted or altered by fluid entering upstream from the IV. This documentation communicates precisely which site was used and why.

#410 · Medical Terminologies and Abbreviations

What do the directional terms medial, lateral, anterior, posterior, superior, and inferior describe?

Answer

Medial means closer to the midline of the body and lateral means farther from it, toward the side. Anterior and posterior describe front and back respectively, and superior and inferior describe above and below.

#411 · Medical Terminologies and Abbreviations

What does CBC stand for and what tube is it collected in?

Answer

CBC stands for complete blood count, one of the most frequently ordered tests in any setting. It is collected in a lavender top tube containing EDTA anticoagulant and evaluates red cells, white cells, and platelets together.

#412 · Medical Terminologies and Abbreviations

What does BUN stand for and what does it assess?

Answer

BUN stands for blood urea nitrogen, a chemistry test that assesses kidney function by measuring a waste product normally filtered out by the kidneys.

#413 · Medical Terminologies and Abbreviations

What does INR stand for, what is it used for, and what tube does it require?

Answer

INR stands for international normalized ratio, a standardized measure of how long blood takes to clot, used heavily to monitor patients on warfarin therapy. It is collected in a light blue top tube with sodium citrate anticoagulant that must be filled to its exact draw volume for the result to be valid.

#414 · Medical Terminologies and Abbreviations

What does PTT stand for?

Answer

PTT stands for partial thromboplastin time, another coagulation study collected in a citrate tube, used to evaluate a different portion of the clotting cascade than the INR.

#415 · Medical Terminologies and Abbreviations

What does HbA1c measure and why does it not require fasting?

Answer

HbA1c refers to glycated hemoglobin, sometimes called hemoglobin A1c, a test that reflects average blood glucose control over roughly the preceding two to three months rather than a single point in time. Because it reflects a longer average rather than a current level, it generally does not require the patient to fast, unlike a fasting glucose.

#416 · Medical Terminologies and Abbreviations

What does TSH stand for, and what does the abbreviation lytes refer to?

Answer

TSH stands for thyroid stimulating hormone, a screening test for thyroid function. Lytes is short for electrolytes, a chemistry panel measuring substances such as sodium, potassium, chloride, and bicarbonate, essential for assessing fluid balance.

#417 · Medical Terminologies and Abbreviations

What does LFTs stand for?

Answer

LFTs stands for liver function tests, a panel of chemistry markers reflecting liver health and function.

#418 · Medical Terminologies and Abbreviations

What does FBS stand for and what patient preparation does it require?

Answer

FBS stands for fasting blood sugar, a glucose test that specifically requires the patient to have had nothing to eat or drink except water for a defined period, usually eight to twelve hours, before the draw. Verifying this preparation is essential before collection.

#419 · Medical Terminologies and Abbreviations

What does C&S; stand for and what is it used for?

Answer

C&S; stands for culture and sensitivity, most often applied to microbiology specimens such as urine or wound swabs. It is used to identify an infecting organism and determine which antibiotics will treat it effectively.

#420 · Medical Terminologies and Abbreviations

What does ESR stand for?

Answer

ESR stands for erythrocyte sedimentation rate, a nonspecific marker of inflammation, collected in a lavender or a dedicated black top tube depending on the method used by the receiving laboratory.

#421 · Medical Terminologies and Abbreviations

What does NPO mean and where does the abbreviation come from?

Answer

NPO comes from the Latin nil per os, meaning nothing by mouth. It appears on orders requiring fasting or on pre-procedural instructions where a patient must not eat or drink anything at all.

#422 · Medical Terminologies and Abbreviations

What does STAT mean on a requisition?

Answer

STAT means immediately, marking an order as urgent. It requires the phlebotomist to prioritize that collection ahead of routine work and deliver the specimen to the laboratory without delay.

#423 · Medical Terminologies and Abbreviations

What does QNS stand for and when is it applied?

Answer

QNS stands for quantity not sufficient, the notation a laboratory applies to a specimen that did not contain enough volume to run the ordered test, often triggering a request for a repeat collection.

#424 · Medical Terminologies and Abbreviations

What does AMA stand for?

Answer

AMA stands for against medical advice, documenting that a patient has chosen to refuse a recommended test, procedure, or admission despite being informed of the risks. A phlebotomist may encounter this directly when a patient declines a scheduled draw.

#425 · Medical Terminologies and Abbreviations

Where is the antecubital area and why is it the preferred venipuncture site?

Answer

The antecubital area is the surface of the arm at the front of the elbow, the site of the median cubital, cephalic, and basilic veins. It is the preferred location for routine venipuncture in most adult patients because these veins are typically large, close to the surface, and well anchored by surrounding tissue.

#426 · Medical Terminologies and Abbreviations

What is a hematoma and what commonly causes it during a draw?

Answer

A hematoma is a localized collection of blood outside the vessel, under the skin, usually caused by blood leaking from the puncture site during or after the draw. It often results from a needle that has gone through the back wall of the vein or from insufficient pressure applied after withdrawal.

#427 · Medical Terminologies and Abbreviations

What is syncope and why must a phlebotomist be prepared for it?

Answer

Syncope refers to a temporary loss of consciousness, commonly known as fainting. It is a real risk during and immediately after venipuncture that every phlebotomist must be prepared to recognize and manage safely.

#428 · Medical Terminologies and Abbreviations

What is the difference between a sclerosed vein and a thrombosed vein?

Answer

A sclerosed vein is one that has become hardened and fibrous, often from repeated punctures over time, and feels firm and cordlike. A thrombosed vein also feels hard, but the underlying cause is a formed clot obstructing the vessel rather than generalized scarring; either finding should prompt selecting a different draw site.

#429 · Medical Terminologies and Abbreviations

What is hemoconcentration and what commonly causes it?

Answer

Hemoconcentration describes a relative increase in the concentration of larger blood components, such as cells and large molecules, relative to plasma. It is most commonly caused by prolonged tourniquet application or by pumping the fist repeatedly before a draw, and it can distort results for several analytes.

#430 · Medical Terminologies and Abbreviations

Why does ISMP Canada discourage the handwritten abbreviation U for units, and what is the deeper lesson for a phlebotomist?

Answer

ISMP Canada, the Institute for Safe Medication Practices Canada, discourages U because, written by hand, it can be mistaken for a zero or a four, turning an intended value into something ten times larger or smaller than what was meant. The deeper lesson is to write any handwritten note on a requisition out clearly and unambiguously rather than relying on personal shorthand someone else might misread later.

#431 · Good Clinical Practice Safety and Infection Control

What is the Canadian term for the infection control foundation applied to every patient encounter without exception?

Answer

Routine practices, which some other countries call standard precautions. Every patient's blood and body fluids are treated as potentially infectious at all times, regardless of the requisition or what the patient reports about their own health.

#432 · Good Clinical Practice Safety and Infection Control

Why can a phlebotomist not safely relax precautions for a patient who denies any risk factors or looks healthy?

Answer

Many bloodborne infections cause no visible symptoms for years, and a recent infection may sit inside a window period where a test comes back negative despite the organism already being present and transmissible. A complete honest history cannot rule out an infection the patient does not yet know they have.

#433 · Good Clinical Practice Safety and Infection Control

What is the first of the four moments of hand hygiene?

Answer

Before patient contact, performed before touching the patient's arm to palpate a vein or apply a tourniquet. This is the earliest point in the encounter where hand hygiene is required.

#434 · Good Clinical Practice Safety and Infection Control

When does the second moment of hand hygiene occur in a venipuncture procedure?

Answer

The second moment is before a clean or aseptic procedure. In phlebotomy this falls just before the needle enters the skin, after site preparation and immediately before venipuncture itself.

#435 · Good Clinical Practice Safety and Infection Control

When does the third moment of hand hygiene occur, and what does it cover?

Answer

The third moment is after a body fluid exposure risk, covering the period right after the draw once the needle is withdrawn, when blood may have contacted the technician's hands or gloves.

#436 · Good Clinical Practice Safety and Infection Control

What is the fourth moment of hand hygiene?

Answer

After patient contact, performed once the encounter is fully finished and the patient has left the immediate collection area.

#437 · Good Clinical Practice Safety and Infection Control

What is the correct technique for using an alcohol-based hand rub?

Answer

Apply enough product to cover all surfaces of both hands, including between the fingers, around the thumbs, and under the nails, and continue rubbing until the hands are completely dry. This typically takes fifteen to twenty seconds.

#438 · Good Clinical Practice Safety and Infection Control

In which two situations must soap and water be used instead of an alcohol-based hand rub?

Answer

When hands are visibly soiled with blood, body fluid, or other contamination, since alcohol does not clean organic material off the skin the way mechanical washing does. Also when caring for a patient under precautions for a spore-forming organism such as C. difficile, since alcohol does not kill spores.

#439 · Good Clinical Practice Safety and Infection Control

Why is alcohol-based hand rub ineffective against C. difficile?

Answer

C. difficile forms spores that are resistant to the killing action of alcohol. Only the physical, mechanical action of washing with soap and running water reliably removes the spores from the skin.

#440 · Good Clinical Practice Safety and Infection Control

What is the absolute minimum PPE standard for every venipuncture or skin puncture?

Answer

Gloves, with no exceptions made for a quick draw, a cooperative patient, or a technician's confidence in their own technique. Gloves must be worn for every single collection performed.

#441 · Good Clinical Practice Safety and Infection Control

Is it acceptable to reuse the same pair of gloves across multiple patients if hand hygiene is performed over top of them between patients?

Answer

No. Gloves must be changed between every patient without exception. Reusing gloves across patients, even with hand hygiene performed over the gloves, is a serious infection control breach.

#442 · Good Clinical Practice Safety and Infection Control

What does each of a gown, a mask, and eye protection guard against in the phlebotomy setting?

Answer

A gown protects clothing and skin from splashes or contact with contaminated surfaces, a mask protects the respiratory tract from droplets, and eye protection such as safety glasses or a face shield protects the mucous membranes of the eyes from splashes.

#443 · Good Clinical Practice Safety and Infection Control

What should a phlebotomist do before entering a patient's room when precaution signage is posted?

Answer

Check the posted precaution signage before entering, gather the appropriate PPE for that precaution level, and don it in the correct sequence before contact, rather than improvising equipment choices at the bedside.

#444 · Good Clinical Practice Safety and Infection Control

What does WHMIS stand for and what version is currently in effect in Canada?

Answer

The Workplace Hazardous Materials Information System, updated to its 2015 version to align with the Globally Harmonized System used internationally. It governs how hazardous chemicals are identified, labelled, and communicated to workers.

#445 · Good Clinical Practice Safety and Infection Control

Name three hazardous products a phlebotomy technician commonly handles that fall under WHMIS.

Answer

Alcohol-based skin antiseptics, chlorhexidine solutions, and surface disinfectants all fall under WHMIS. Certain adhesive removers can also be included.

#446 · Good Clinical Practice Safety and Infection Control

What is a Safety Data Sheet, abbreviated SDS, and what information does it provide?

Answer

A standardized document that accompanies every hazardous product used in a workplace, providing detailed information on its composition, physical and health hazards, safe handling and storage requirements, and the correct first aid response if a worker is exposed. Every workplace must keep SDS documents accessible to staff.

#447 · Good Clinical Practice Safety and Infection Control

What do WHMIS hazard pictograms look like and what is their purpose?

Answer

Standardized black-and-white symbols inside a red diamond border, printed directly on product labels. They give an immediate visual warning of a specific hazard class, such as flammable or corrosive, before a product is even opened.

#448 · Good Clinical Practice Safety and Infection Control

Why does isopropyl alcohol require special caution in a home care collection setting?

Answer

Isopropyl alcohol, the antiseptic used on nearly every venipuncture site, is itself flammable and should never be used near an open flame or a source of ignition, which matters when drawing near a lit stove during a home visit.

#449 · Good Clinical Practice Safety and Infection Control

What is the absolute rule regarding a used needle in Canadian occupational health and safety practice?

Answer

No recapping, no bending, and no breaking of a used needle, under any circumstance. Recapping in particular is a leading cause of preventable needlestick injuries because the two-handed capping motion places the non-dominant hand in the path of the sharp point.

#450 · Good Clinical Practice Safety and Infection Control

Where must a used needle be disposed of, and what does this mean for how sharps containers are positioned?

Answer

Immediately into a puncture-resistant sharps container, directly at the point of use. The sharps container must come to the needle rather than the needle travelling to the container, so a technician should never carry an exposed used needle across a room in search of a disposal point.

#451 · Good Clinical Practice Safety and Infection Control

What should a phlebotomist do if a portable collection tray has no sharps container within immediate reach of the draw site?

Answer

The draw should not proceed until a sharps container is available within immediate reach. Collection must never begin without a compliant disposal point already in place at the point of use.

#452 · Good Clinical Practice Safety and Infection Control

What is the first step in the correct response to a needlestick or sharps injury?

Answer

Immediate wound care: wash the affected area with plain soap and water, allowing the water to flow over the wound. The wound should never be squeezed, milked, or sucked, since neither action reduces infection risk and both can cause further tissue trauma or contamination.

#453 · Good Clinical Practice Safety and Infection Control

After wound care, what is the second step following a sharps injury, and why is timing important?

Answer

Immediate reporting to a supervisor or the designated occupational health contact, done as soon as possible after the injury rather than at the end of a shift. Timing affects the options available for post-exposure management.

#454 · Good Clinical Practice Safety and Infection Control

What does the third step of the sharps injury response protocol involve?

Answer

Prompt follow-up through occupational health services, which may include assessing the source patient's infection status where possible, baseline testing of the exposed worker, and consideration of post-exposure prophylaxis when indicated.

#455 · Good Clinical Practice Safety and Infection Control

A technician is unsure whether a needlestick incident was significant enough to report. What should they do?

Answer

Always report it. The decision about clinical significance belongs to occupational health, not to the technician's own judgment in the moment.

#456 · Good Clinical Practice Safety and Infection Control

What PPE do contact precautions require, and give two organisms that commonly need them.

Answer

Contact precautions require a gown and gloves for anyone entering the patient's space, along with careful hand hygiene on exit. MRSA and C. difficile are common examples of organisms requiring contact precautions.

#457 · Good Clinical Practice Safety and Infection Control

What PPE do droplet precautions require, and what organism is a commonly cited example?

Answer

A surgical mask along with eye protection when working at close range with the patient, since venipuncture places the technician's face close to the patient. Influenza is a commonly cited example requiring droplet precautions.

#458 · Good Clinical Practice Safety and Infection Control

What PPE and room conditions do airborne precautions require, and which organisms are classic examples?

Answer

A fit-tested N95 respirator rather than a surgical mask, and care to enter and exit through a room kept under negative pressure so air flows into the room rather than out. Tuberculosis and measles are the classic examples requiring airborne precautions.

#459 · Good Clinical Practice Safety and Infection Control

Why does wearing only a surgical mask into a room requiring airborne precautions leave a technician unprotected?

Answer

A surgical mask does not filter the fine airborne particles that organisms like tuberculosis and measles can remain suspended in and travel on. Only a fit-tested N95 respirator provides the filtration needed, so the mismatch is a real protection failure, not merely a technical non-compliance.

#460 · Good Clinical Practice Safety and Infection Control

How should used gauze, alcohol swabs, or a tourniquet visibly soiled with blood be discarded?

Answer

Into a designated biohazard bag, typically identifiable by its yellow or orange colour and the biohazard symbol printed on it. This applies to materials contaminated with blood or body fluids that are not sharp.

#461 · Good Clinical Practice Safety and Infection Control

What two problems arise from mixing biohazardous waste into the general waste stream?

Answer

The infection control problem is that untrained ordinary waste handlers may be exposed to contamination they had no reason to expect. The regulatory problem is that biomedical waste disposal is governed by provincial and territorial regulation, and mixing streams is a direct violation of those requirements.

#462 · Good Clinical Practice Safety and Infection Control

What is contact time, also called dwell time, when disinfecting a surface such as a phlebotomy chair armrest?

Answer

The length of time a disinfectant must remain visibly wet on a surface, as specified by the manufacturer, for many hospital-grade disinfectants running from one to several minutes. A surface wiped dry before this time has elapsed has not actually been disinfected, even though it looks clean.

#463 · Good Clinical Practice Safety and Infection Control

Should a splash of blood that contacts intact skin, rather than a mucous membrane, be reported as an occupational incident?

Answer

Yes. The professional and legal obligation is to report every exposure, injury, or near-miss promptly and accurately, even one that seems minor, rather than deciding independently that it was too minor to matter.

#464 · Good Clinical Practice Safety and Infection Control

Why does underreporting of incidents harm the workplace as a whole, beyond the individual technician?

Answer

Incident data drives safety improvements. Patterns of needlestick injuries clustered around a particular device, procedure, or time of shift can only be identified and corrected if every incident, including near-misses, is captured in the reporting system.

#465 · Good Clinical Practice Safety and Infection Control

Give two examples of ergonomic risks in phlebotomy and how correct positioning helps prevent them.

Answer

Bending awkwardly over a patient's arm, twisting the wrist repeatedly during needle insertion, and standing in a static position for long periods all contribute to cumulative strain injuries. Adjusting the height of the phlebotomy chair or stool and repositioning oneself relative to the patient protects the technician's long-term physical health.

#466 · Site Preparation

Why should a phlebotomist avoid drawing through or proximal to a hematoma?

Answer

Blood pooled in the tissue from a prior puncture or injury dilutes and alters the specimen's composition, producing an inaccurate result. The opposite arm should be selected if possible; if no alternative exists and the draw cannot wait, a site distal to the hematoma may be used, but never through it or proximal to it.

#467 · Site Preparation

Why is an edematous limb avoided for venipuncture?

Answer

Edema is swelling caused by excess fluid accumulating in the tissue, and this fluid dilutes the blood at that site, meaningfully skewing laboratory results. A clearer site elsewhere on the body should be selected in favour of the edematous limb.

#468 · Site Preparation

What venipuncture precaution applies to an arm on the side of a mastectomy with lymph node removal?

Answer

Removal of axillary lymph nodes disrupts normal lymphatic drainage, leaving the arm prone to lymphedema and at elevated risk of infection. Canadian clinical practice generally directs phlebotomists to draw from the unaffected arm instead, when a suitable alternative exists.

#469 · Site Preparation

What should a phlebotomist do when a patient has had bilateral mastectomy with node dissection on both sides?

Answer

There is no clearly unaffected arm to default to, so the phlebotomist should consult the ordering clinician or follow the facility's documented escalation procedure rather than making an independent judgment call. This respects the altered physiology of both limbs instead of treating one arbitrarily as usable.

#470 · Site Preparation

What is the rule for venipuncture on an arm with a dialysis fistula or graft?

Answer

Never perform venipuncture on that arm without specific specialized training and explicit authorization. A fistula is a surgically created artery-to-vein connection, and a graft uses synthetic tubing for the same purpose; both are lifelines for kidney failure patients that a misplaced needle can permanently damage.

#471 · Site Preparation

What should a phlebotomist do if asked to draw from a fistula or graft arm with no other site offered?

Answer

Escalate to a supervisor rather than proceeding. Standard phlebotomy training does not qualify a technician to draw from a fistula or graft arm, since a misplaced needle or unnecessary puncture can damage or destroy this surgically established, difficult-to-replace vascular access.

#472 · Site Preparation

How should a phlebotomist handle a patient with a running intravenous line when both arms are compromised?

Answer

The opposite arm should be used whenever available; if both arms are compromised, an unavoidable IV-side draw may be permitted under strict facility protocol. A site distal to the IV insertion point is selected, the IV may be paused briefly by qualified nursing staff, and a discard volume of blood is collected first to clear the line of infusion fluid.

#473 · Site Preparation

Why must an IV-side blood draw be documented as such?

Answer

The infused fluid can contaminate the specimen and skew results for analytes such as electrolytes and glucose, even after a discard volume is drawn. Documenting the draw as IV-side ensures the receiving laboratory and clinician can correctly interpret results that may still be affected.

#474 · Site Preparation

Why should burns, extensive scarring, tattoos, and areas of extensive bruising be avoided as venipuncture sites?

Answer

Damaged or altered skin heals unpredictably, may harbour higher bacterial loads, and can make both antiseptic preparation and vein visualization unreliable. A different, undamaged site should be selected whenever one is available.

#475 · Site Preparation

Describe the correct antiseptic technique for a routine venipuncture site.

Answer

The site is cleansed with a 70 percent isopropyl alcohol swab using firm circular friction, starting at the intended puncture point and moving outward in expanding circles, never returning the swab back over an already-cleaned area. The alcohol is then allowed to air dry completely, typically thirty to sixty seconds, before the needle is inserted.

#476 · Site Preparation

Why should a phlebotomist never blow on, fan, or wipe an alcohol-prepped site with gauze before inserting the needle?

Answer

Doing so defeats the purpose of the antiseptic by disturbing it before it has had time to work, and can also introduce contaminants or cause a stinging sensation for the patient. The alcohol must be left to air dry on its own, typically for thirty to sixty seconds.

#477 · Site Preparation

Why do blood cultures require a more rigorous antiseptic protocol than routine venipuncture?

Answer

A blood culture detects bacteria in the bloodstream, and if skin flora are inadvertently introduced into the culture bottle during collection, the laboratory may report a false-positive result. A false positive can lead to unnecessary antibiotic treatment, additional invasive testing, and extended hospitalization from a contaminant that was never in the patient's blood.

#478 · Site Preparation

What antiseptic agents are commonly used for blood culture site preparation?

Answer

A chlorhexidine-based antiseptic is commonly used, or in some facilities a two-step iodine-then-alcohol protocol. Either is applied in a center-outward circular motion with a longer contact and dry time than routine alcohol preparation, often a full thirty seconds to two minutes depending on the product and facility policy.

#479 · Site Preparation

What is the rule about re-palpating a site after it has been prepared for a blood culture?

Answer

The site should not be re-palpated with a bare or ungloved finger. If additional palpation is genuinely necessary, a sterile gloved finger that has itself gone through the antiseptic process must be used, or the site should be re-cleansed if palpation disturbs it.

#480 · Site Preparation

In what clinical situations is capillary puncture indicated instead of venipuncture?

Answer

Capillary collection is used when a patient has very small, fragile, or difficult-to-access veins, when the patient is a very young infant for whom a venous draw's volume and risk would be disproportionate, for many point-of-care tests such as glucose monitoring, and for patients on strict fluid restriction.

#481 · Site Preparation

What are the correct puncture sites for an infant heel stick?

Answer

The medial or lateral plantar surface of the heel, the fleshy portions on the inner and outer edges of the sole near the heel. The curvature at the very back of the heel, along with any area of prior puncture, must never be used.

#482 · Site Preparation

What is the maximum recommended puncture depth for an infant heel stick, and why?

Answer

Generally not exceeding approximately two millimetres, because puncturing too deeply risks striking the calcaneus, the heel bone, which lies closer to the skin surface in an infant's small foot than intuition might suggest. Striking bone can cause osteomyelitis, a serious bone infection.

#483 · Site Preparation

Why is warming the heel recommended before an infant heel stick?

Answer

Warming, typically with a commercial heel warmer or a warm, moist cloth for a few minutes, dilates the local capillaries and improves blood flow. This allows an adequate sample to be collected with fewer punctures and less squeezing of the tissue.

#484 · Site Preparation

What is the correct fingerstick site for capillary collection in older children and adults?

Answer

The palmar surface of the middle or ring finger, on the lateral, or side, aspect of the fingertip rather than dead center on the pad, where nerve endings are more concentrated and tissue is thinner over bone. The thumb and the smallest finger are avoided.

#485 · Site Preparation

Why is fingerstick capillary collection avoided in newborns?

Answer

A newborn's fingers are too small for safe capillary collection by this method, mirroring the restriction against using the curved back of the heel. The heel stick is the appropriate capillary technique for this age group instead.

#486 · Site Preparation

Why should a cold, cyanotic, or edematous finger not be used for a fingerstick?

Answer

Poor peripheral circulation will yield an inadequate or unrepresentative sample. Warming the hand beforehand can help improve circulation and make an otherwise unsuitable finger usable.

#487 · Site Preparation

Why is the first drop of blood wiped away after a fingerstick puncture?

Answer

The first drop is often diluted with residual tissue fluid or alcohol from the antiseptic preparation and does not accurately reflect the patient's blood composition. It is wiped away with clean gauze before the actual sample is collected.

#488 · Site Preparation

How does aging skin affect venipuncture technique in geriatric patients?

Answer

Skin thins and loses collagen and elastic support with age, becoming considerably more fragile and prone to both bruising and tearing, particularly where tape or a bandage is applied and later removed. Care must be taken when removing adhesive, since aggressive removal can tear fragile skin.

#489 · Site Preparation

Why are geriatric veins often described as fragile and rolling, and how should a phlebotomist compensate?

Answer

They have lost some of the surrounding tissue support that normally anchors them in place, so they tend to shift away from the needle during insertion rather than staying put. The vein should be anchored firmly with a thumb positioned below the intended puncture site to counteract rolling.

#490 · Site Preparation

What tourniquet adjustment is recommended for geriatric venipuncture?

Answer

Gentler tourniquet tension should be used, and the tourniquet should be released promptly. This, combined with allowing extra time for the entire procedure, accounts for the fragility of aging skin and veins.

#491 · Site Preparation

What anatomical challenge does obesity present for venipuncture, and what adjustment is appropriate?

Answer

Veins that would normally be superficial and easily palpable may instead be considerably deeper, obscured beneath a thicker layer of subcutaneous adipose tissue, making both visualization and palpation more difficult. A longer needle may be selected when appropriate to reach a deeper vein safely, while standard careful palpation technique otherwise still applies.

#492 · Site Preparation

What is blind probing, and why is it never appropriate?

Answer

Blind probing means repeatedly redirecting a needle beneath the skin in search of a vein that cannot be felt or confirmed. It increases the risk of nerve injury, arterial puncture, hematoma formation, and unnecessary patient discomfort, so an alternate site or colleague assistance should be sought instead.

#493 · Site Preparation

Why do oncology and other chronically ill patients often have fragile veins?

Answer

Repeated venipuncture over the course of a long illness, along with the effects of certain chemotherapy agents on vein walls, can leave veins scarred, hardened, or prone to collapsing under the vacuum pressure of a standard collection tube. This makes vein selection and technique especially important in this population.

#494 · Site Preparation

What should a phlebotomist do when a chronically ill patient has a central venous line or implanted port?

Answer

Recognize that the device exists and coordinate with the patient's care team rather than defaulting to an unnecessary separate peripheral venipuncture. Draws from these devices are performed by specially trained staff, typically nursing, following device-specific protocols.

#495 · Site Preparation

When a peripheral draw is genuinely required in an oncology patient with fragile veins, what precautions reduce complication risk?

Answer

Choosing the most robust remaining vein, using a smaller-gauge needle if the vein's condition warrants it, and applying extra care with tourniquet pressure all reduce the risk of vein collapse or hematoma. These adjustments respect how compromised the patient's remaining veins already are.

#496 · Site Preparation

What is vasovagal syncope in the context of a blood draw?

Answer

A sudden drop in heart rate and blood pressure triggered by the stress or sight of the procedure. It is one of the more common patient safety events in phlebotomy, and recognizing its early warning signs is a core competency.

#497 · Site Preparation

What are the early warning signs that a patient may be about to faint during a blood draw?

Answer

Pallor, a visible loss of colour in the face, sweating that appears suddenly without an obvious cause, reported dizziness or lightheadedness, and complaints of nausea are all early indicators. Any of these should prompt immediate action to protect the patient.

#498 · Site Preparation

What should a phlebotomist do if a patient shows signs of vasovagal syncope during collection?

Answer

Stop the procedure immediately if the sample has not yet been fully collected, or complete it as quickly as safely possible if it is nearly finished. The patient should then be repositioned to reduce fall risk: lying flat if already on an examination table, or lowering the head between the knees if seated with no bed available.

#499 · Site Preparation

How long should a patient who has experienced vasovagal syncope be monitored before leaving the collection area?

Answer

Continuously, until they have fully recovered with normal colour returned and symptoms resolved, before being permitted to sit up fully or leave unattended. A patient allowed to walk away too soon risks a secondary fall injury that can be far more serious than the original blood draw.

#500 · Site Preparation

Why does the certification exam frequently test site preparation and special populations through scenario-style questions?

Answer

This content sits at the intersection of anatomy, infection control, patient identification, and communication, so a single scenario can require applying several chapters' worth of knowledge at once. A perfectly executed technique on the wrong site or with the wrong protocol still represents a broken link in the chain of safe, accurate phlebotomy.